

OIL-AT-RISK: A FRAMEWORK FOR STUDYING OIL PRICE VULNERABILITIES

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Abstract

Crude oil prices act as a key macroeconomic variable, often serving as a primary input in global inflation dynamics, industrial production costs, and international trade balances. The propagation of geopolitical tensions through energy markets can generate heightened volatility, which carries macroeconomic consequences and potential systemic risks for policymakers, firms, and institutional investors. Many traditional econometric approaches assume that the average historical effect of a geopolitical shock adequately represents the underlying stochastic process. However, in the context of crude oil prices and geopolitical shocks, where data tend to exhibit substantial heteroskedasticity, asymmetric responses, and influential tail observations that can distort linear estimates, this assumption may prove overly restrictive. To address the limitations of conditional mean forecasting, the present research adapts the *Growth-at-Risk* framework to model the distribution of future

crude oil price growth. Rather than focusing solely on average baseline effects, the analysis estimates how fluctuations in geopolitical risk shift specific quantiles of the return distribution. This approach aims to capture heterogeneity across outcomes and to describe both downside and upside market risks. The empirical strategy relies on the European Brent crude oil spot price and a dictionary-based Geopolitical Risk Index. Drawing on daily returns and a set of control variables, including a realised-volatility proxy, the U.S. dollar index, and a shipping-cost measure, the study specifies multiple quantile regressions to isolate the tail-risk imprint of geopolitical uncertainty at a one-trading-week horizon. A backtesting framework combining the Kupiec unconditional-coverage test and the dynamic quantile test evaluates the conditional risk measures against realised outcomes, and an entropy-based indicator is examined for early-warning content. The full analytical pipeline is implemented in a self-contained, open-source Python module to promote replicability.

KEYWORDS: Geopolitical Risk, Brent Crude Oil, Quantile Regression, Growth-at-Risk, Tail Risk, Volatility, Johnson S_U , CAViaR, Expected Shortfall, Kullback–Leibler Divergence.

JEL CODES: C21, C22, C53, C58, F51, G17, Q43.

1 Introduction

The study of geopolitical risk and its cascading effects on crude oil prices remains a significant concern for the global economy. While inflation dynamics vary due to a multitude of complex macroeconomic factors, oil prices serve as one of their primary drivers, often influencing the cost of global industrial production and affecting international trade balances and current account deficits. Geopolitical shocks, ranging from localised military conflicts and diplomatic standoffs to sweeping international embargoes and maritime blockades, tend to propagate rapidly through energy markets. These events can introduce profound uncertainty that may prompt immediate adjustments in consumption, investment, and precautionary storage among economic agents. Understanding not only whether geopolitical risk affects oil prices, but how exposed future prices might be to the extreme tails of these global risks, is therefore a valuable endeavour for economic stabilisation and financial risk management.

Standard macroeconomic and econometric frameworks have historically tended to favour modelling the conditional mean of asset prices and macroeconomic indicators. By frequently relying on Ordinary Least Squares (OLS) regression techniques, classical studies often estimate the average historical effect of a geopolitical shock, treating variance as a relatively constant dispersion around that mean. However, the observable dynamics of global oil markets frequently challenge this setup. Energy market data commonly exhibit intense volatility clustering, asymmetric shock responses, and heavy-tailed distributions. In this case, the average effect may be skewed by extremes without adequately capturing them, which can leave policymakers and risk managers with a misleading sense of both the magnitude and the likelihood of adverse price movements.

This empirical challenge aligns with an extensive literature in energy econometrics regarding the potential non-linearities of the oil market. As argued by Hamilton (2003), modelling oil shocks often benefits from a flexible approach to account for the possibility that price increases may have a different macroeconomic impact than price decreases. Kilian and Vigfusson (2011) provide evidence suggesting that the macroeconomic response to energy shocks can be asymmetric. Consequently, a strictly linear methodology risks understating downside risks. The exposure of the market may not be adequately captured by mean forecasts; it typically requires a framework specifically equipped to describe asymmetric impacts.

Geopolitical shocks introduce an additional layer of non-linear complexity. Pinchetti (2024) suggests that large geopolitical shocks raise Brent prices and inflation expectations by triggering precautionary wait-and-see behaviours and a surge in speculative demand for crude. This is further contextualised by Blanchard and Galí (2010), who highlight how modern oil shocks interact differently with the macroeconomy than they did in the 1970s, mediated by factors such as wage rigidities, central bank credibility, and the shifting

energy intensity of the economy. The physical materialisation of a threat is not required to damage the economy: Kilian et al. (2026) find that an unanticipated increase in the probability of a major geopolitical oil production shortfall, a perceived disaster scenario, can be sufficient to generate a marked surge in oil prices and macroeconomic uncertainty. The *Growth-at-Risk* methodology, adapted here as *Oil-at-Risk*, offers an econometric solution suited to this dynamic. By isolating specific quantiles, it describes how surges in geopolitical uncertainty fatten the tails of the distribution, allowing for a nuanced analysis of tail behaviour that linear models frequently miss.

To quantify these tail risks conditional on changes in geopolitical and macroeconomic regressors, the empirical procedure is structured in sequential steps. Rather than collapsing the complexity of the oil market into a single conditional mean forecast, the analysis begins by estimating multiple quantile regressions at selected points of the conditional distribution, from the lower tail through the median to the upper tail, so that the model can reveal how geopolitical risk influences both the median and the extremes. The raw output from the quantile regressions provides a discrete inverse cumulative distribution function, which is then smoothed by fitting a flexible continuous distribution. Once the continuous conditional distribution is built, the final step evaluates the magnitude of the upside and downside risks, describing how exposed the baseline forecast might be to unexpected shocks. This exposure is measured using Value-at-Risk, Expected Shortfall, Kullback–Leibler divergence, and distributional skewness, each capturing a distinct dimension of the risk profile.

The remainder of this thesis is organised as follows. Section 2 reviews the relevant literature. Section 3 sets out the contributions. Section 4 describes the data. Section 5 develops the methodological framework, covering quantile regression, the direct forecasting design, the CAViaR augmentation, distributional fitting via the Johnson S_U , and the suite of risk and vulnerability metrics. Section 6 presents the empirical implementation and out-of-sample results. Section 7 studies the entropy-based early warning indicator. Section 8 concludes. The appendices collect the full technical development of every diagnostic and evaluation test, and a Graphical Annex displays the figures referenced throughout.

2 Background and Literature Review

The methodological foundation of this research builds upon the Vulnerable Growth paradigm introduced by Adrian et al. (2019). By modelling the conditional distribution of gross domestic product growth as a function of economic and financial conditions, their framework observed that deteriorating financial conditions tend to increase conditional volatility and shift the lower quantiles downward, while upper quantiles remain comparatively stable. This paradigm indicated that downside risks exhibit significant variance based on financial tightness, highlighting the limitations of conditional mean estimation

and the value of evaluating tail risks. Extending this rationale suggests that similar non-linear, tail-heavy dynamics might govern other critical macroeconomic variables that are frequently subjected to sudden, severe shocks.

The recent literature has adapted this tail-risk perspective to the effects of geopolitical uncertainty on energy markets. Brignone et al. (2025) examine the non-linear transmission mechanisms of geopolitical risk shocks, suggesting that large positive shocks can trigger a disproportionate amplification channel driven by rising uncertainty. By decomposing geopolitical risk, they observe that geopolitical acts often function similarly to negative demand shocks, whereas geopolitical threats tend to behave as positive uncertainty shocks that elevate oil prices and inflation expectations. Similarly, Verduzco-Bustos and Zanetti (2026) use high-frequency data to isolate geopolitical oil price shocks, finding that these events typically resemble severe supply disruptions, inducing immediate production declines and sharp price increases. They further note that these shocks are coupled with heightened uncertainty that prompts a distinct precautionary inventory response, characterised by an initial surge in inventories followed by a gradual drawdown as the immediate geopolitical threat recedes. Their work reinforces the view that geopolitical risk does not merely shift the conditional mean; it reshapes the distribution of future oil price outcomes.

The quantile regression methodology was proposed by Koenker and Bassett (1978), who introduced the estimation of conditional quantile functions as an alternative to the conditional mean. Their approach provides a distribution-agnostic estimator that is robust to outliers and capable of revealing heterogeneous covariate effects across the conditional distribution. In the context of risk management, quantile regression acquired a dynamic dimension through the Conditional Autoregressive Value-at-Risk (CAViaR) framework of Engle and Manganelli (2004), which allows the conditional quantile to depend autoregressively on its own past values and on past violations, thereby capturing the persistence and clustering of tail events that static models miss.

The distributional fitting step of the *Growth-at-Risk* framework has traditionally relied on the Azzalini skewed- t distribution, as in Adrian et al. (2019). However, the choice of parametric family is not innocuous. The Johnson S_U distribution, introduced by Johnson (1949), offers an alternative with independent control over skewness and kurtosis through two shape parameters, and a closed-form quantile function that eliminates the need for numerical root-finding. These properties make it suited to modelling commodity returns, where the direction and magnitude of tail asymmetry shift as geopolitical conditions evolve. The S_U family has been used in portfolio management and Value-at-Risk contexts, and its application to the distributional step of a *Growth-at-Risk*-type framework is, to our knowledge, uncommon in the energy-finance literature.

The backtesting of conditional risk measures draws on a well-established body of work. Kupiec (1995) proposed the unconditional coverage test that compares the empirical

frequency of Value-at-Risk exceedances against the nominal rate. The dynamic quantile test of Engle and Manganelli (2004) provides a stringent evaluation by testing whether the hit sequence is a martingale difference sequence, orthogonal to its own past. For Expected Shortfall, the backtesting framework of Acerbi and Székely (2014, 2017) exploits the joint elicibility of VaR and ES, constructing tests based on the severity of exceedances rather than merely their frequency. The scoring-rule theory of Gneiting and Raftery (2007) provides the grounding for strictly proper scoring rules used in model comparison. Together, these tools form the backtesting apparatus applied in this thesis. The information-theoretic component of the framework rests on Shannon (1948) and the relative-entropy measure of Kullback and Leibler (1951).

3 Contributions

This thesis makes the following contributions to the literature on commodity market risk and econometric methodology.

First, it introduces the concept of *Oil-at-Risk* (OaR), adapting the *Growth-at-Risk* (GaR) framework of Adrian et al. (2019) to the crude oil market. The adaptation enables probabilistic tail-risk assessments of oil prices in the same language used for macrofinancial surveillance.

Second, it integrates Conditional Autoregressive Value-at-Risk (CAViaR) ideas (Engle and Manganelli, 2004) into the OaR framework. This enrichment supplements the static quantile regression step of the canonical GaR procedure with variables that encode the model’s own recent boundary violations, aiming to capture the persistence and asymmetric adjustment of tail risk observed in oil price behaviour.

Third, it departs from the conventional use of the skewed- t distribution for distributional fitting by adopting the Johnson S_U (JSU) family (Johnson, 1949). The JSU system provides a four-parameter characterisation capable of matching a range of skewness and kurtosis combinations. The skewed- t remains the reference in the canonical *Growth-at-Risk* framework (Adrian et al., 2019); it is not estimated here, and no head-to-head comparison between the two families is attempted.

Fourth, it examines an entropy-based early warning indicator that exploits daily data. The indicator is the Kullback–Leibler divergence (Kullback and Leibler, 1951) of the fitted conditional density against a reference density, decomposed by tail. Its anticipatory content is assessed with the Cross-Correlation Function and the Granger causality test (Granger, 1969), rather than being assumed.

Fifth, it applies a common backtesting battery uniformly across all model variants. Kupiec unconditional coverage tests (Kupiec, 1995) and Diebold–Mariano comparisons of predictive accuracy (Diebold and Mariano, 1995) are applied to every specification, with

the dynamic quantile test (Engle and Manganelli, 2004) used to assess the hit-sequence dynamics of the CAViaR-augmented models.

Sixth, it delivers a self-contained, open-source Python module that implements the full OaR pipeline, covering data preprocessing, quantile regression under both standard and CAViaR-augmented specifications, distributional fitting, entropy computation, and backtesting diagnostics. The module is publicly available in the following repository at <https://github.com/Alejandro-bot-one/Oil-at-Risk> to support replication and extension.

4 Data Description

The dependent variable is the European Brent crude oil spot price (USD per barrel, U.S. Energy Information Administration via FRED), sampled daily from 2 January 1990 to 30 April 2026, a window spanning numerous major geopolitical crises (Figure 1). Brent is the standard global benchmark, reflecting international seaborne market conditions. Prices are transformed into daily log-returns, and weekends and market holidays are removed to keep a continuous trading-day alignment. The sample median daily log-return is close to zero, consistent with the near-martingale behaviour of high-frequency commodity returns; this matters below, because it implies the median quantile carries little directional signal and most of the conditioning information is expected in the tails.

The main explanatory variable is the daily Geopolitical Risk Index (GPRD), built by Caldara and Iacovello (2018) as a dictionary-based measure of the share of newspaper coverage devoted to adverse geopolitical events. It is entered as a seven-day moving average, matching the calendar-week basis on which the index is published (Section 5.2); country-specific versions allow localised risk origins to be examined.

Three exogenous controls isolate the geopolitical effect. The U.S. Dollar Index (log-differenced) captures the dollar–oil link: since Brent is priced in dollars, an appreciation tends to depress its nominal price, and controlling for it prevents exchange-rate moves from being read as geopolitical shocks. Realised volatility controls for volatility clustering and heteroskedasticity, separating the structural pricing of geopolitical threat from general market turbulence; its handling differs between the in-sample and out-of-sample exercises (Sections 5.2 and 5.4). The Baltic Dry Index (log-differenced), the cost of shipping bulk raw materials by sea, proxies global industrial demand and controls for demand-side pressures largely free of speculative noise.

5 Methodology

When examining the effect of geopolitical risk on the rate of change of Brent crude prices, traditional linear methodologies reveal their limitations. An initial look at the data

displays a high presence of influential observations, data points located far outside the dense central cloud that can heavily distort OLS regressions. Because OLS minimises the sum of squared residuals, it is highly sensitive to these extreme outliers, which inflate the variance of the estimates and can bias the slope in either direction, so that the fitted mean relationship offers little reliable predictive power in the tails. Linear methods focus entirely on the expected value or conditional mean, ignoring parts of the distribution such as the tails. The *Growth-at-Risk* methodology, adapted here as *Oil-at-Risk*, provides a direct econometric alternative. By isolating specific quantiles, it describes how surges in geopolitical uncertainty fatten the tails of the distribution, enabling a nuanced analysis of tail behaviour that linear models miss.

5.1 Quantile Regression

While standard OLS regression provides a powerful tool for estimating the Conditional Expectation Function, it operates under strict assumptions and is highly sensitive to outliers. Quantile regression, introduced by Koenker and Bassett (1978), estimates the Conditional Quantile Function (CQF). Instead of modelling the mean, it models a specific quantile τ (for instance, the 10th or 90th percentiles), making it distribution-agnostic and robust to extreme outliers.

The CQF, given a vector of regressors X_i , is the inverse of the conditional cumulative distribution:

$$Q_\tau(Y_i | X_i) = F_Y^{-1}(\tau | X_i). \quad (1)$$

Just as OLS derives the conditional mean by minimising the expected squared error, quantile regression derives the CQF by minimising an expected asymmetric loss:

$$Q_\tau(Y_i | X_i) = \arg \min_{q(X_i)} \mathbb{E}[\rho_\tau(Y_i - q(X_i))], \quad (2)$$

where $\rho_\tau(u)$ is the check function. To trace out specific percentiles of a distribution, the regression must penalise under-predictions and over-predictions asymmetrically. If we are estimating the 90th quantile ($\tau = 0.90$), the model should heavily penalise under-predictions (negative residuals) and lightly penalise over-predictions (positive residuals). To achieve this while maintaining the convexity required for optimisation, a piecewise linear function based on the absolute error is constructed. Multiplying the positive piece by the target quantile τ and the negative piece by $(1 - \tau)$ introduces the necessary asymmetry. The check function is written as:

$$\rho_\tau(u) = u \cdot (\tau - \mathbf{1}_{u < 0}), \quad (3)$$

where $\mathbf{1}_{u < 0}$ is an indicator function taking the value 1 if the residual is negative and 0

otherwise. Defining the residual as $u_t = y_t - x_t' \beta_\tau$, the objective is to find the parameter vector $\hat{\beta}_\tau$ that minimises the sum of this asymmetric check function across all observed periods:

$$\hat{\beta}_\tau = \arg \min_{\beta_\tau \in \mathbb{R}^k} \sum_{t=1}^T \rho_\tau(y_t - x_t' \beta_\tau). \quad (4)$$

The economic appeal of this estimator is immediate. In the context of crude oil markets, OLS would estimate a single slope linking geopolitical risk to average returns. By contrast, quantile regression estimates a separate slope at each quantile level, revealing whether geopolitical shocks predominantly compress the lower tail (raising downside risk), stretch the upper tail (amplifying upside potential), or shift the entire distribution. This heterogeneity across quantiles is the identifying premise of the *Oil-at-Risk* framework. The optimisation procedure, which transforms the non-differentiable check function into a standard linear programme, is detailed in Appendix A.

5.2 Empirical Specification

The empirical specification retained for in-sample estimation regresses daily Brent crude oil log-returns on a seven-day moving average of the Geopolitical Risk Index and a vector of control variables. For each quantile in the target set $\tau \in \{0.01, 0.05, 0.25, 0.50, 0.75, 0.95, 0.99\}$, the empirical equation is:

$$Q_\tau(\Delta \text{Brent}_t) = \beta_0(\tau) + \beta_1(\tau) \text{GPRD-MA7}_t + \gamma(\tau)' X_t, \quad (5)$$

where the in-sample control vector is:

$$X_t = (\Delta \text{USDI}_t, \text{RV}_{t-1}, \Delta \text{BDI}_t)'. \quad (6)$$

A single specification is imposed uniformly across all quantiles to preserve comparability across the conditional distribution and to prevent tail-specific overfitting. The estimated coefficient paths across τ , together with the OLS benchmark, are displayed in Figure 3.

The primary regressor is a seven-day moving average of the Geopolitical Risk Index, chosen to match a calendar week of the underlying index, which is compiled on a calendar-week basis (seven calendar days):

$$\text{GPRD-MA7}_t = \frac{1}{7} \sum_{j=0}^6 \text{GPRD}_{t-j}. \quad (7)$$

The seven-day moving average is used to capture the transmission of geopolitical risk through the news cycle, which propagates to market positioning over several days rather than within a single session, so that $\hat{\beta}_1(\tau)$ reflects sustained geopolitical stress rather than the transient spike of an individual headline. The seven-day length is a deliberately simple,

arbitrary choice made on this rationale; it is not optimised, and alternative smoothing windows are entirely admissible and would be a natural robustness dimension for future work.

Realised volatility enters the in-sample regression at lag $t - 1$ rather than contemporaneously, as a deliberate precaution against look-ahead bias. Because realised volatility at time t is computed from intraperiod price movements, it is partly determined by the dependent variable itself, which would induce a simultaneity problem in a contemporaneous ($h = 1$) regression. Using RV_{t-1} ensures the regressor is fully observable before period t opens. This concern is specific to the in-sample specification, in which the target and the volatility proxy share the same date. In the out-of-sample direct forecasting exercise of Section 5.4, where the target is the return realised h trading days ahead, contemporaneous realised volatility RV_t is already part of the information set available at the forecast origin and cannot leak future information; the lag is therefore unnecessary there, and RV_t is used.¹

Equation (5) represents a parsimonious specification selected from a broader set of candidate models evaluated during the model-selection stage. It is presented primarily as a descriptive benchmark to illustrate how geopolitical risk affects different regions of the conditional distribution of Brent returns while controlling for the main sources of contemporaneous market conditions. Although alternative specifications were also considered during model development, equation (5) provides a framework that serves as a natural starting point for the forecasting methodology developed below. In Section 5.4, this baseline specification is adapted to the direct forecasting setting, where the same conditioning variables are used to model the conditional distribution of future Brent returns.

Finally, the candidate specifications were also screened for multicollinearity using standard diagnostic measures (see, for instance, Table 1), and no evidence of problematic collinearity was found among the retained regressors.

¹Equivalently: in the in-sample equation the target y_t and RV_t share the same date, so entering RV_t would place a function of the dependent variable on the right-hand side and induce simultaneity; the one-period lag removes it. In the direct-forecasting equation the target is y_{t+h} with $h \geq 1$, so RV_t is dated strictly before the target and belongs to the origin- t information set, carrying no look-ahead. The two specifications thus use RV_{t-1} and RV_t respectively for the same underlying reason.

Table 1: Contemporaneous Pearson correlations among the control variables and the Brent log-return over the estimation sample. ΔUSDI is the log-difference of the U.S. Dollar Index, RV the realised-volatility proxy, ΔBDI the log-difference of the Baltic Dry Index, and ΔBrent the Brent log-return.

	ΔUSDI	RV	ΔBDI	ΔBrent
ΔUSDI	1.0000	0.0145	-0.0129	-0.2285
RV	0.0145	1.0000	-0.0177	0.0821
ΔBDI	-0.0129	-0.0177	1.0000	0.0120
ΔBrent	-0.2285	0.0821	0.0120	1.0000

5.3 Diagnosis of the Quantile Regression

The diagnostic battery provides broad empirical support for the validity of specification (5). The Koenker–Machado $R^1(\tau)$ (Koenker and Machado, 1999) indicates that the inclusion of the regressors yields a meaningful reduction in the weighted objective function relative to the intercept-only benchmark, with explanatory power concentrated in the tail quantiles, precisely where the *Oil-at-Risk* metric is most informative. This concentration is expected rather than anomalous: the conditioning variables are designed to describe extreme events, and, because the median daily return is close to zero and carries little directional signal, the regressors add proportionally more at the tails than at the centre. The QAR unit root test of Koenker and Xiao (2004) rejects the null of quantile-specific non-stationarity at all targeted quantiles, consistent with mean-reverting behaviour of Brent log-returns throughout the conditional distribution. The Wald test for equality of slopes across quantiles rejects the null of coefficient constancy, supporting the view that the regressors exert heterogeneous effects across the conditional distribution, which is the identifying premise of the *Oil-at-Risk* framework. No quantile crossing violations are detected in the fitted paths, consistent with the monotonicity requirement across all observations in the sample. The complete technical development of each diagnostic, with its null hypothesis, test statistic, and asymptotic distribution, is provided in Appendix B.

5.4 Direct Multi-Step Forecasting

The out-of-sample analysis is genuinely predictive: it targets the conditional distribution of Brent returns h periods ahead using only information available today. The iterated approach—chaining a one-step model forward—is unavailable here, because conditional quantiles do not obey a law of iterated expectations: a multi-step quantile cannot be recovered by composing one-step quantile forecasts the way a conditional mean can.

The analysis therefore uses direct forecasting: a separate quantile regression is estimated for each horizon h by pairing the information available at t with the realised future value

Y_{t+h} ,

$$\hat{\beta}^{(h)}(\tau) = \arg \min_{\beta} \sum_{t=1}^{T-h} \rho_{\tau}(Y_{t+h} - X'_t \beta), \quad Q_{\tau}(Y_{t+h} | X_t) = X'_t \beta^{(h)}(\tau). \quad (8)$$

Because a dedicated model is fitted at each horizon, the estimator targets the horizon directly and avoids the compounding bias of the recursive scheme; re-estimating over a grid of h also shows how the geopolitical risk premium propagates over time. This is also the setting in which contemporaneous realised volatility RV_t can be used without look-ahead bias (Section 5.2), since the target lies h days beyond the forecast origin.

Estimation and evaluation are kept strictly apart: at each origin the model trains on the rolling window $[t - 1000, t)$ (about four trading years), issues one h -step forecast, and advances one day, sweeping every eligible origin. The realised returns against the forecasted quantile envelope are shown in Figure 4, and the in-sample forecast around a stressed episode in Figure 10.

5.5 Competing Specifications: Capturing Geopolitical Tail Risk

The standard quantile regression is an accurate description of the typical conditional quantile, but it is, in a precise sense, memoryless about its own failures. It maps today's geopolitical and financial covariates into a quantile forecast and then forgets whether that forecast was breached. In tranquil markets this is harmless; in the tails it is not.

The economic object of interest is not the centre of the distribution but its extreme edge, the threshold beyond which an oil position suffers a large loss. Extreme oil moves are not independent draws: they cluster. A supply shock threatening the Strait of Hormuz, or a demand collapse of the COVID-19 type, does not resolve in a single session; it propagates through inventory decisions, futures-curve repositioning, and risk re-pricing over days and weeks. Empirically this shows up as volatility clustering in the tail, where a Value-at-Risk breach today materially raises the probability of another breach tomorrow. A model that treats each tail event as an isolated surprise is therefore caught off guard precisely when accuracy matters most.

This is the gap that Engle and Manganelli (2004) address with the CAViaR framework, in which the conditional quantile is allowed to depend autoregressively on its own past behaviour, so that the tail remembers recent stress. The present thesis does not estimate a full CAViaR recursion; instead it imports the central insight into the direct-forecasting design by augmenting the standard specification with variables that encode the model's own recent boundary violations. Two augmentation variants are considered.

5.5.1 Model 2: CAViaR Indicator Augmentation

The first variant appends a pair of binary stress flags to the regressor set. Let $\hat{Q}_{s|s-h}^{\text{high}}$ and $\hat{Q}_{s|s-h}^{\text{low}}$ denote the upper and lower boundary quantiles (the 0.95 and 0.05 marks) forecast for date s from information dated $s - h$. The flags evaluate whether the realised return

breached that pre-committed boundary:

$$\text{upside_breach}_t = \mathbf{1}(y_t > \hat{Q}_{t|t-h}^{\text{high}}), \quad (9)$$

$$\text{downside_breach}_t = \mathbf{1}(y_t < \hat{Q}_{t|t-h}^{\text{low}}). \quad (10)$$

Both flags are dated t and depend only on quantities observable by t : the realised return y_t and a boundary that was itself fixed at $t - h$. When these regressors enter the forecast of the future target y_{t+h} formed at origin t , every ingredient is therefore dated at or before t , strictly earlier than the target, so no future information is used.² A positive coefficient on these indicators means that the model widens its predicted quantile corridor after a recent breach, absorbing the persistence that a memoryless specification would miss.

5.5.2 Model 3: CAViaR Severity Augmentation

A binary flag records that a breach happened but is blind to its violence: a 1% breach and a 10% breach speak with the same voice. Yet for tail risk the magnitude is much of the story, the difference between a manageable loss and a blow-up. The severity variant therefore replaces the indicators with the absolute size of the exceedance, the fallout error:

$$\text{upside_severity}_t = \max(0, y_t - \hat{Q}_{t|t-h}^{\text{high}}), \quad (11)$$

$$\text{downside_severity}_t = \max(0, \hat{Q}_{t|t-h}^{\text{low}} - y_t), \quad (12)$$

Entering the fallout error in absolute value forces its coefficient to act as a pure stress-intensity loading, uncontaminated by the sign of the exceedance, so the upper- and lower-tail severities read as independent measures of how hard the market overshot. Because both augmentations need a realised value to define a breach, the stress signal they carry is sharp at very short horizons but decays as h lengthens, an economic limitation the out-of-sample results make concrete.

5.5.3 The Dynamic Quantile Test and the Role of Breach Memory

The motivation for these augmentations is made precise by the Dynamic Quantile (DQ) test of Engle and Manganelli (2004), the sharpest available check on whether a quantile model has absorbed the dynamics of its own violations. Define the centred hit sequence

$$\text{Hit}_t(\tau) = \mathbf{1}(y_t < \hat{q}_t(\tau)) - \tau, \quad (13)$$

which has mean zero under correct specification because a well-calibrated τ -quantile is breached exactly a fraction τ of the time. Correct *dynamic* specification requires more

²Concretely, the regressor vector at origin t contains breach_t , which is a function of y_t and of the boundary $\hat{Q}_{t|t-h}$ committed h days earlier; the object being predicted is y_{t+h} . The availability lag between the most recent breach information and the target is thus at least one period and the flag can never read the realised value it is helping to forecast.

than the right average: it requires that a breach today be unpredictable from anything known yesterday, so that the hit sequence is a martingale difference with respect to the information set Ω_{t-1} ,

$$H_0 : \mathbb{E}[\text{Hit}_t(\tau) \mid \Omega_{t-1}] = 0. \quad (14)$$

The test operationalises this by projecting the hits onto an instrument matrix Z_t built from a constant, L lagged hits $\text{Hit}_{t-1}(\tau), \dots, \text{Hit}_{t-L}(\tau)$, and the lagged fitted quantile $\hat{q}_{t-1}(\tau)$, and testing the joint significance of the projection through the Wald form

$$DQ(\tau) = \frac{\widehat{\text{Hit}}' Z (Z' Z)^{-1} Z' \widehat{\text{Hit}}}{\tau(1 - \tau)} \xrightarrow{d} \chi_q^2, \quad (15)$$

with q equal to the column dimension of Z ; the equivalent Lagrange-multiplier form $N \cdot R^2$ from the auxiliary regression converges to the same law. Rejection means the violation process is still predictable, i.e. the model is caught off guard by clustered tail events. The two CAViaR augmentations attack exactly the term the DQ test interrogates: by feeding the model's own lagged breach information back into the quantile forecast, they are constructed so that recent violations no longer carry predictive information for the next one. In that sense the CAViaR-augmented specifications pass the DQ test by construction, since they internalise precisely the lagged-hit dependence that the statistic is designed to detect.

5.6 Distribution Fitting: From Discrete Quantile Forecasts to a Continuous Conditional Density

The direct forecasting model delivers, for every forecast origin t and horizon h , a finite set of predicted conditional quantiles:

$$\hat{Q}_{t+h} = \left\{ \hat{q}_{t+h}(\tau_1), \dots, \hat{q}_{t+h}(\tau_K) \right\}, \quad \tau_k \in \{0.01, 0.05, 0.25, 0.50, 0.75, 0.95, 0.99\}. \quad (16)$$

These point forecasts are the discrete skeleton of the conditional distribution of future Brent returns. A portfolio manager who needs only a single Value-at-Risk threshold at the 5th percentile can read $\hat{q}_{t+h}(0.05)$ directly and stop. But we would like to extend the *Oil-at-Risk* framework well beyond a single threshold.

Consider what a central bank or a sovereign wealth fund actually needs when a geopolitical crisis unfolds in the Persian Gulf. The question is not merely whether the 5th percentile of the return distribution has moved; it is how the entire shape of the conditional density has changed. Has the left tail fattened while the right tail remained stable, signalling a pure downside tail risk? Has skewness reversed, as it does when a demand-side collapse overtakes a supply-side fear? To compute Expected Shortfall, one must integrate below the quantile. To measure Kullback–Leibler divergence between

successive conditional densities, one must have a density. To study how geopolitical shocks reshape the skewness and kurtosis of the forecast distribution across horizons, one must recover the full shape of the curve, not merely a handful of points on it. The discrete quantile skeleton must therefore be interpolated into a continuous parametric density that preserves the information encoded in the quantile forecasts while supplying the analytic tractability that downstream tail-risk metrics require.

5.6.1 The Johnson S_U Distribution

The discrete quantile skeleton is interpolated with the Johnson S_U distribution, a four-parameter family $\Theta = (\xi, \lambda, \gamma, \delta)$ obtained by applying an inverse-hyperbolic-sine transformation to a standard normal. It is chosen for two reasons. First, its two shape parameters give independent control over skewness (γ) and tail weight (δ), so it can track the asymmetric tail rotations of crude oil returns, right-tail fattening under supply threats, left-tail fattening under demand collapses, and it nests the Gaussian in the limit $\gamma \rightarrow 0$, $\delta \rightarrow \infty$. Second, its cumulative distribution function admits a closed-form inverse, so the quantile function can be evaluated without numerical root-finding, which makes the Minimum Distance Estimation step below as fast as a least-squares problem. The formal definition and quantile function are given in Appendix D.

5.6.2 Minimum Distance Estimation

In the classical Maximum Likelihood setting, the analyst observes many realisations from the return-generating process, each an independent draw from the stochastic process whose density is to be estimated. The likelihood surface is well identified, the sample is large, and the estimator is efficient. The setting that arises after the quantile regression step is different. For a given forecast origin t and horizon h , we do not observe a large sample of future returns. We observe a small vector of K predicted quantiles, one for each targeted quantile level. With $K = 7$, we have seven data points, not seven thousand. The natural estimator in this setting is Minimum Distance Estimation (MDE), a technique that estimates the parameters of a distribution by minimising a measure of discrepancy between its theoretical quantile function and a set of target quantiles (here, the predicted quantiles), rather than by maximising a likelihood over raw observations. Concretely, the MDE finds the parameter vector $\Theta = (\xi, \lambda, \gamma, \delta)$ that makes the theoretical quantile function of the chosen distribution pass as close as possible to the predicted quantiles:

$$\hat{\Theta}_t = \arg \min_{\Theta} \sum_{k=1}^K [\hat{q}_{t+h}(\tau_k) - Q_{JSU}(\tau_k; \Theta)]^2. \quad (17)$$

5.6.3 Evaluation: The Probability Integral Transform

Once the MDE has produced a fitted conditional distribution for every forecast origin, the critical question becomes whether those distributions are accurate descriptions of

the future reality they purport to model. The foundational diagnostic is the Probability Integral Transform (PIT), formalised in the context of density forecast evaluation by Diebold et al. (1998). For each forecast origin t in the out-of-sample window, the PIT evaluates the realised return y_{t+h} under the conditional CDF that was forecasted for that date:

$$u_t = F(y_{t+h} \mid \hat{\Theta}_t). \quad (18)$$

If the fitted distribution is the true conditional distribution, the transformed value u_t is exactly Uniform(0, 1):

$$P(U \leq u) = u \quad \text{for all } u \in [0, 1]. \quad (19)$$

The contrapositive supplies the diagnostic: if the sequence of PITs departs from uniformity, the forecasted distribution is misspecified. Departures manifest in characteristic ways that carry direct economic meaning. An excess of PITs near zero indicates that the model's left tail is too thin, understating the probability of large negative returns, precisely the kind of miscalibration that would lead a risk manager to understate her Expected Shortfall during a geopolitical crisis. The PIT histogram and the corresponding diagonal (empirical-versus-theoretical CDF) plot are shown in Figures 12 and 13.

The formal assessment is provided by the Kolmogorov–Smirnov (KS) test (Massey, 1951), which measures the supremum absolute deviation between the empirical CDF of the PIT sequence and the theoretical CDF of the Uniform(0, 1) distribution:

$$D_n = \sup_{u \in [0, 1]} |\hat{F}_n(u) - u|. \quad (20)$$

The PIT evaluation is conducted over the same out-of-sample window used for the coverage backtests. This design constitutes a single, continuous goodness-of-fit assessment that spans the entire out-of-sample period, evaluating every fitted density across every forecast origin simultaneously. It is more demanding than a pointwise coverage backtest, because it asks whether the entire shape of the conditional density, its location, its spread, its skewness, and its tail weight, is correctly specified at every date. A PIT sequence consistent with uniformity is therefore a necessary condition for treating the fitted distributions as a credible probabilistic characterisation of future oil price tail risks under geopolitical duress.

5.7 Risk and Vulnerability Metrics

Having recovered a continuous conditional Johnson S_U density at every forecast origin, the framework extracts the risk and vulnerability metrics that constitute the diagnostic output of the *Oil-at-Risk* framework. A density, by itself, is not a decision tool. A risk manager facing a potential supply disruption in the Strait of Hormuz does not consult a probability density function; she asks how large her worst-case loss might be, how severe the tail would be if that worst case materialises, and whether the current distribution

has shifted in a way that signals an abnormal accumulation of downside risk relative to historical norms. These are different questions, and they require different metrics.

5.7.1 Value-at-Risk

Value-at-Risk is the α -quantile of the conditional return distribution. For the left tail:

$$\text{VaR}_{t+h}^{\text{Left}}(\alpha) = Q_{\alpha}(r_{t+h} \mid \hat{\Theta}_t), \quad (21)$$

and for the right tail:

$$\text{VaR}_{t+h}^{\text{Right}}(1 - \alpha) = Q_{1-\alpha}(r_{t+h} \mid \hat{\Theta}_t), \quad (22)$$

where $\text{VaR}^{\text{Left}} < 0$ represents a loss threshold and $\text{VaR}^{\text{Right}} > 0$ represents a gain threshold. In plain terms, the left-tail measure reports the return the market would realise under a very adverse scenario, that is, a return so low that it is expected to be exceeded on the downside only with probability α ; the right-tail measure reports the return under a very favourable scenario, so high that it is expected to be exceeded on the upside only with probability α . Computing both is essential because the conditional distribution fitted by the Johnson S_U is generically asymmetric: when the Geopolitical Risk Index is elevated and the dollar is weakening, the two tails need not move in parallel, and reporting only the left tail would discard the information contained in the right tail's response to the same conditioning variables.

The conditional VaR follows directly from the fitted density:

$$\text{VaR}_{t+h}^{\text{Cond, Left}}(\alpha) = \hat{\xi}_t + \hat{\lambda}_t \sinh\left(\frac{\Phi^{-1}(\alpha) - \hat{\gamma}_t}{\hat{\delta}_t}\right). \quad (23)$$

A useful observation is that no time-scaling is needed for the conditional VaR. The quantile regression that produced the inputs to the distribution fitting step was estimated with the target variable y_{t+h} , the return over the next h trading days. The fitted Johnson S_U distribution is therefore already a distribution of h -step-ahead returns. Its α -quantile is directly the h -day VaR, without any square-root-of-time adjustment.

The unconditional VaR, computed by historical simulation over a rolling window of $W = 1,000$ observations, serves as the benchmark against which the conditional measure is assessed. When the two VaR series track each other closely, the conditioning variables are adding little information beyond what the recent history already contains. When they diverge, the conditional model is detecting a shift in conditions that the backward-looking historical measure has not yet absorbed. The conditional and unconditional VaR and CVaR series are displayed in Figure 15 and 16.

5.7.2 Expected Shortfall

Value-at-Risk answers how bad a loss can be at a given confidence level, but it is silent on what happens beyond that threshold. Two distributions can share the same 97.5th percentile while having radically different tail behaviour beyond it: one may decay rapidly to zero, implying that exceedances beyond the VaR are rare and mild, while the other may exhibit a heavy tail extending far into the loss region, implying that when the VaR is breached, the resulting loss is likely to be large. In crude oil markets, this distinction is not hypothetical. The distribution of returns during the COVID-19 demand collapse of April 2020 had a left tail extending to losses of 30 percent or more in a single day, while the distribution during the preceding months of gradual price decline had a much thinner left tail despite a similar VaR at the 2.5th percentile.

Expected Shortfall addresses this limitation by measuring the average loss in the tail beyond the VaR threshold:

$$\text{ES}_{t+h}^{\text{Left}}(\alpha) = \mathbb{E}\left[r_{t+h} \mid r_{t+h} \leq \text{VaR}_{t+h}^{\text{Left}}(\alpha)\right]. \quad (24)$$

The left-tail Expected Shortfall is always more negative than the left-tail VaR, because the conditional expectation in the tail lies beyond the tail's boundary by construction. The difference $\text{ES}^{\text{Left}} - \text{VaR}^{\text{Left}}$ measures the expected severity of a VaR exceedance, a quantity of direct interest to a risk manager who has already accepted that the VaR threshold may be breached and wants to know how bad the outcome is likely to be.

Expected Shortfall enjoys a property that Value-at-Risk does not: it is a coherent risk measure in the sense of Artzner et al. (1999). The critical property is sub-additivity, which was one of the principal motivations behind the Basel Committee's decision, in the Fundamental Review of the Trading Book, to replace VaR with Expected Shortfall as the basis for internal-model capital calculations under Basel III (Basel Committee on Banking Supervision, 2019).

When the conditional density is available in closed form, the Expected Shortfall integral is evaluated by numerical integration on a fine grid of 5,000 points spanning the support of the distribution:

$$\text{ES}_{t+h}^{\text{Cond, Left}}(\alpha) \approx \frac{\sum_{x_i \leq \text{VaR}_{t+h}^{\text{Left}}} x_i f(x_i \mid \hat{\Theta}_t) \Delta x}{\sum_{x_i \leq \text{VaR}_{t+h}^{\text{Left}}} f(x_i \mid \hat{\Theta}_t) \Delta x}. \quad (25)$$

5.7.3 Kullback–Leibler Divergence as a Vulnerability Indicator

Value-at-Risk and Expected Shortfall are point statistics read off the conditional density at specific tail probabilities; they do not measure how far the whole density has moved from

a reference state. Kullback–Leibler divergence (Kullback and Leibler, 1951) supplies that global measure of distributional distance. It is non-negative, zero only when the two densities coincide, and asymmetric in an economically meaningful way: it penalises a reference that treats as near-impossible the large losses the conditional forecast deems plausible—the dangerous case for a risk manager. Following Adrian et al. (2019), it is decomposed into left- and right-tail components at the tail mark $\tau = 0.95$ and measured against two baselines: a Normal density (a kurtosis-sensitive early-warning signal of departure from Gaussianity) and the unconditional Johnson S_U fitted by Maximum Likelihood (Figure 11), against which a high divergence signals a genuine shift in the conditional distribution rather than mere non-normality. The formal definition, its information-theoretic interpretation, and the tail-decomposition integrals are given in Appendix D.1.

5.7.4 Distributional Skewness

The Fisher skewness coefficient of the fitted Johnson S_U distribution is computed analytically using the closed-form expressions derived by Johnson (1949):

$$\gamma_1 = -\frac{\sqrt{\omega}(\omega + 2) \left[3 \sinh\left(\frac{\gamma}{\delta}\right) + \sinh\left(\frac{3\gamma}{\delta}\right) \right]}{4 \left[\frac{\omega-1}{2} \left(\omega \cosh\left(\frac{2\gamma}{\delta}\right) + 1 \right) \right]^{3/2}}, \quad (26)$$

where $\omega = \exp(1/\delta^2)$. The skewness is negative when the left tail is heavier than the right, signalling an asymmetric accumulation of downside risk, and positive when the right tail dominates. The economic content of time-varying skewness in crude oil markets is substantial. During periods of geopolitical escalation, the right tail fattens relative to the left and skewness becomes positive or less negative. During demand-side recessions, the opposite pattern emerges. Tracking conditional skewness over time reveals the direction in which the balance of risks is tilted at each forecast origin.

5.8 Computational Configuration

The risk metrics are computed out-of-sample at every trading day in the test span, from the same rolling pipeline.

Two out-of-sample spans appear in the thesis, by design rather than inconsistency. The forecast-evaluation backtests of Section 6 (pinball, fallout, Kupiec, Diebold–Mariano) are reported over 2 January 2023 to 23 April 2026, 864 origins for the standard model and 517 for the CAViaR variants, a window chosen to keep the coverage counts comparable across the three specifications. The conditional density, its risk metrics, and the entropy series of Section 7 are evaluated over the longer 21 June 2012 to 24 April 2026 span (2,168 daily values; 2,167 after differencing). Both are genuine out-of-sample objects from the same pipeline, and each table states its own sample.

The forecast horizon is $h = 5$ trading days (one trading week), applied consistently

across the forecasting, risk-metric, and early-warning steps.

Two probability levels operate in parallel and should not be conflated. The quantile forecasts and their coverage and pinball evaluation are assessed at $\tau = 0.95$ (the 0.05 mark), while the risk measures read from the fitted density-, VaR and Expected Shortfall, are quoted at 97.5% confidence (the 0.025 tail), following the Basel III Fundamental Review of the Trading Book (Basel Committee on Banking Supervision, 2019). The quantile regression is estimated on the grid $\tau \in \{0.01, 0.05, 0.25, 0.50, 0.75, 0.95, 0.99\}$, and the 2.5th percentile is read from the fitted Johnson S_U density.

The historical VaR and ES use a rolling window of $W = 1,000$ observations (about four trading years), matching the training window; the quantile-regression coefficients are refitted every 30 out-of-sample days and cached in between, so the density still updates daily through the changing covariates while the coefficient mapping is held fixed between refits.

6 Empirical Implementation and Out-of-Sample Results

6.1 Evaluating the Forecasts

Three complementary lenses judge the forecasts: a strictly proper loss measured over the rolling window, a magnitude-of-exceedance audit, and a formal unconditional-coverage backtest. The mechanics of every test are derived in full in Appendix E; here the treatment is kept brief and focused on what each test certifies. All figures below are computed at the operative horizon $h = 5$ and, where a tail is specified, at $\tau = 0.95$.

6.1.1 The Pinball Loss and the Generalisation Gap

A quantile cannot be scored with the symmetric mean-squared error, because over- and under-prediction must be penalised by different amounts. The asymmetric pinball (check) loss supplies that asymmetry:

$$L_\tau(e) = e \cdot (\tau - \mathbf{1}_{e < 0}), \quad e = Y_{t+h} - \hat{y}_{t+h|t}(\tau). \quad (27)$$

It is the strictly proper scoring rule for quantiles (Gneiting and Raftery, 2007): it is minimised if and only if the forecast equals the true conditional quantile, so a lower out-of-sample pinball loss is strong evidence of a better tail estimate. Tracking the out-of-sample pinball loss across a dense grid of horizons for each specification reveals how predictive accuracy degrades as the horizon lengthens: a loss profile that rises sharply with the horizon is the classic signature of overfitting, whereas a flat profile indicates that the model is capturing structural signal. These per-horizon loss profiles are shown in Figure 5. In the present experiment the benchmark’s loss stays broadly flat across the horizons

examined, whereas the CAViaR variants deteriorate at longer horizons, consistent with their stale-information handicap.

6.1.2 Fallout (Severity) Audit

Coverage frequency alone can hide catastrophic single-day failures, so the magnitude of every breach is also audited: how far the realised return overshoot the predicted 1%–99% tunnel. For risk management, frequent but tiny breaches are far preferable to rare but massive ones, since the latter betray a structural failure to price an impending shock. At $h = 5$ the three models keep the realised return inside the predicted envelope on 97.83% (Standard), 97.17% (CAViaR-indicator) and 97.26% (CAViaR-severity) of test days, with no isolated catastrophic overshoot. The full breach paths for the three models, including the behaviour of the tunnel during the 2020 crisis, are collected in Figures 6,7 and 8.

6.1.3 Coverage Backtests

Because the model’s output is an estimated quantile that functions as a Value-at-Risk threshold, its adequacy is judged with a risk-management backtest built on the binary hit sequence. The Kupiec (1995) Proportion-of-Failures test checks whether the empirical hit rate $\hat{\pi} = x/T$ matches the nominal τ through a likelihood-ratio statistic that is asymptotically χ_1^2 :

$$LR_{UC} = -2 \ln \left[\frac{(1 - \tau)^{T-x} \tau^x}{(1 - \hat{\pi})^{T-x} \hat{\pi}^x} \right] \sim \chi_1^2, \quad \hat{\pi} = \frac{x}{T}. \quad (28)$$

The test evaluates unconditional coverage only: it asks whether breaches occur at the correct *frequency*, and is silent about their timing. This is a deliberate choice in the present setting. Because the $h = 5$ direct forecasts overlap, the hit sequence is serially dependent by construction (an $MA(h - 1)$ structure), which distorts tests that model the *dependence* of breaches, but leaves the expected breach *count*, and hence the unconditional-coverage statistic, unaffected.³ Table 2 summarises the results, and the corresponding coverage-convergence plot for the main models is shown in Figure 9.

³The overlap inflates the variance of the breach count but not its mean, so the Kupiec point assessment of frequency is robust to the induced serial dependence, whereas an independence-based conditional-coverage test applied naively to the same overlapping sequence would be size-distorted. Conditional-coverage testing is therefore not reported here.

Table 2: Kupiec unconditional-coverage backtest at $\tau = 0.95$, $h = 5$. $\hat{\pi}$ is the realised share of days on which the return fell below the forecast (target 0.95); LR_{UC} is the Kupiec likelihood-ratio statistic and p_{UC} its p -value. All models fail to reject correct unconditional coverage at the 5% significance level.

Model	Test window	N	$\hat{\pi}$	LR_{UC} (p_{UC})
Standard QReg	2023-01-02 – 2026-04-23	864	0.9387	2.189 (0.139)
CAViaR-indicator	2023-01-02 – 2026-04-23	864	0.9381	1.437 (0.231)
CAViaR-severity	2023-01-02 – 2026-04-23	864	0.9342	2.471 (0.116)

6.2 Model Selection and Forecast Horizons

The Diebold–Mariano test (Diebold and Mariano, 1995) arbitrates between the three models. Using the asymmetric tick loss and a HAC variance with bandwidth $h - 1$, which matches the $MA(h - 1)$ autocorrelation of h -step direct forecast errors, it tests the null of equal predictive accuracy. All three specifications are estimated and evaluated over the same data and the same test span; the test statistic is formed from the paired loss differential on the forecast origins common to the models being compared, so the comparison is like-for-like even though the CAViaR variants consume a larger number of initial rows when constructing their breach regressors, which is why their reported window carries fewer origins. At $\tau = 0.95$, $h = 5$, Table 3 presents the results.

Table 3: Pairwise Diebold–Mariano tests, tick loss, $\tau = 0.95$, $h = 5$. No pairwise comparison is significant even at the 10% level.

Comparison	α	t -stat	p -value	Verdict
Standard vs CAViaR-i	+0.00901	+0.909	0.364	no sig. difference
Standard vs CAViaR-s	−0.09372	−1.081	0.280	no sig. difference
CAViaR-i vs CAViaR-s	−0.10273	−1.183	0.237	no sig. difference

On the headline predictive-accuracy criterion the three models are statistically indistinguishable: at the operative horizon $h = 5$ the Diebold–Mariano test finds no significant difference in tick loss for any pairwise comparison, so no model dominates. The average tick loss, the only proper score for a quantile, is 0.2380 (Standard), 0.2290 (CAViaR-indicator) and 0.3317 (CAViaR-severity); the indicator augmentation is marginally lower and the severity augmentation marginally higher than the benchmark, but neither gap is statistically significant.

Because no model dominates at the operative horizon $h = 5$, and because the overarching objective of this thesis is to forecast the full conditional distribution of Brent returns across a span of horizons, the standard quantile-regression direct forecast is retained as the production model: it is the most parsimonious of the three, it passes the unconditional-coverage backtest, and it is statistically tied with the CAViaR variants on the only proper

score. The CAViaR augmentations are not discarded on principle. Their breach regressors carry the sharpest signal when the forecast horizon is short enough, so at horizons below $h = 5$ a CAViaR specification may or may not overtake the benchmark.

Table 4: Loss-minimising horizon per model and quantile (lower average pinball loss is better). Entries read $h^* \rightarrow$ loss.

Model	$\tau = 0.05$	$\tau = 0.50$	$\tau = 0.95$
Standard QReg	$h = 2 \rightarrow 0.3017$	$h = 4 \rightarrow 0.9375$	$h = 3 \rightarrow 0.2856$
CAViaR-indicator	$h = 9 \rightarrow 0.2960$	$h = 6 \rightarrow 0.9742$	$h = 7 \rightarrow 0.2806$
CAViaR-severity	$h = 20 \rightarrow 0.3397$	$h = 6 \rightarrow 0.9573$	$h = 16 \rightarrow 0.3810$

Table 5: Evidence summary across the three direct-forecasting models at $\tau = 0.95$, $h = 5$.

Test	What it certifies	Standard	CAViaR-i	CAViaR-s
Pinball / gen. gap	Proper-scoring fit, no overfit	Pass	Pass	Pass
Fallout audit	No catastrophic overshoot	97.83%	97.17%	97.26%
Kupiec POF	Correct breach frequency	Pass (0.139)	Pass (0.231)	Pass (0.116)
Diebold–Mariano	Relative accuracy	baseline	n.s.	n.s.

6.3 Backtesting the Risk Measures

Risk measures are only as credible as the evidence that they perform as advertised. A Value-at-Risk at the 97.5 percent confidence level is, implicitly, a promise: in no more than 2.5 percent of periods should the realised loss exceed the stated threshold. If exceedances occur materially more often, the model is understating risk and the capital reserves it implies are insufficient. If exceedances are materially rarer, the model is overly conservative and the capital it reserves is unnecessarily large, imposing an economic cost in the form of foregone investment.

The framework evaluates the VaR measure with the Kupiec unconditional-coverage test and the Expected Shortfall measure with a joint-elicibility severity test. The Kupiec test counts the number of times the realised return exceeds the VaR threshold and tests whether this count is consistent with the nominal exceedance probability $\alpha = 1 - \text{confidence}$:

$$LR_{UC} = -2 \ln \left[\frac{\alpha^x (1 - \alpha)^{n-x}}{\hat{\pi}^x (1 - \hat{\pi})^{n-x}} \right], \quad \hat{\pi} = \frac{x}{n}. \quad (29)$$

As in the forecast-evaluation backtests, only unconditional coverage is assessed, because the overlapping h -step forecasts induce serial dependence in the hit sequence that would distort an independence-based conditional-coverage test but leaves the frequency assessment intact.

For Expected Shortfall, the Acerbi and Székely (2014, 2017) tests exploit the joint elicibility of VaR and ES, constructing test statistics based on the severity of VaR

exceedances rather than merely their frequency. This furnishes a direct check on the depth of the tail beyond the VaR threshold, the quantity that ES is designed to summarise and that a frequency-based VaR backtest cannot see.

Together, these procedures evaluate both the threshold accuracy (VaR) and the tail-severity accuracy (ES) of the *Oil-at-Risk* framework’s risk measures. The complete derivation, hypotheses, and test statistics for every backtest are collected in Appendix E.

7 Early Warning System

7.1 Motivation: From Vulnerability Measurement to Anticipation

The risk metrics of the preceding section describe the conditional distribution at each date; the Kullback–Leibler divergence, in particular, measures how far it has moved from its reference state. The question this section asks is whether that movement is predictive: if a spike in tail entropy tends to precede large oil price movements, the measure works not as a coincident diagnostic but as an early warning signal, giving risk managers a window to adjust hedges and margins before stress materialises rather than after.

The hypothesis is tested with two independent tools: the Cross-Correlation Function, to see whether entropy leads returns (its peak correlation falling at a positive lag), and the Granger causality test, to check whether lagged entropy predicts returns beyond the information already in their own past.

7.2 Choice of the Early Warning Indicator

Building the indicator requires three choices: the reference baseline, the estimation window, and the forecast horizon.

The baseline is the Normal density, rather than the unconditional Johnson S_U used by Adrian et al. (2019). The unconditional S_U fitted to Brent returns is near-symmetric in its centre but strongly heavy-tailed (excess kurtosis of order ten; Figure 11), so the KL divergence against a Normal reference acts as a clean detector of tail fattening: near zero in calm periods, rising sharply when the tails widen under geopolitical or demand-side shocks. Both references are recalibrated on the data available at each forecast origin, so the series stays out-of-sample. The two entropy series are shown in Figures 14 and 15.

The window is rolling, of $W = 1,000$ observations (about four trading years), rather than expanding: it reacts faster to changing conditions, is cheaper to update in real time, and matches the window used for the VaR and Expected Shortfall computations.

The horizon is $h = 5$ trading days, matching the forecasting and risk-metric steps: shorter horizons leave too little time to act, while longer ones make the quantile regression, and hence the fitted density, noisier.

The indicator is therefore the rolling-window KL divergence of the conditional Johnson

S_U density against the Normal baseline at $h = 5$, decomposed at the tail mark $\tau = 0.95$ into full-distribution, left-tail (downside), and right-tail (upside) components.

7.3 The Cross-Correlation Function

The first diagnostic is the Cross-Correlation Function (CCF), a standard tool for detecting lead-lag relationships between time series (Bujosa et al., 2013; Brockwell and Davis, 1991). It reports the correlation between the indicator at time t and the target at time $t + h$, under the convention that $h > 0$ means the indicator leads the target by h periods. If the entropy series leads Brent returns, the CCF peaks at a positive lag h^* , which is then the anticipation horizon; a peak at $h^* = 0$ (coincident) or $h^* < 0$ (lagging) would be of no use for early warning.

Significance at each lag is judged against Bartlett’s 95 percent band, $\pm 1.96/\sqrt{N}$, under the null of independence (Bartlett, 1946). The estimated function is shown in Figure 17.

Only the entropy indicators are first-differenced before the CCF is computed; the Brent return enters in levels. The return is already stationary (a log-difference of prices), so differencing it again would over-difference the series and attenuate the lead-lag correlations of interest, whereas first-differencing the entropy series removes its persistent level shifts during prolonged stress while preserving its lead-lag dynamics.

7.4 Granger Causality

The CCF identifies the lag at which the correlation between two series is maximised, but correlation does not establish predictive content in a rigorous econometric sense. Two series can be highly correlated at a positive lag because they are both driven by a common third factor, without the leading series carrying any incremental predictive information beyond what the target’s own history already provides. The Granger causality test (Granger, 1969) addresses this limitation by asking a more precise question: do lagged values of the entropy indicator improve the forecast of Brent returns beyond the forecast that can be constructed from the returns’ own lagged values alone?

The test is implemented by comparing two nested linear models. The restricted model regresses the target series on its own lagged values:

$$Y_t = \alpha_0 + \sum_{j=1}^p \alpha_j Y_{t-j} + e_t, \quad (30)$$

where Y_t denotes the Brent return at date t in levels (the series is stationary by construction), p is the lag order, α_0 is an intercept, and e_t is the error term. The unrestricted model augments this specification with lagged values of the entropy indicator:

$$Y_t = \alpha_0 + \sum_{j=1}^p \alpha_j Y_{t-j} + \sum_{j=1}^p \beta_j X_{t-j} + u_t, \quad (31)$$

where X_{t-j} denotes the first-differenced entropy indicator at lag j , and $\beta_1, \dots, \beta_p$ are the coefficients on the lagged indicator values. The null hypothesis is that the indicator carries no incremental predictive content:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0. \quad (32)$$

Under this null, the lagged entropy values are jointly irrelevant for predicting future returns once the returns' own history is accounted for. Rejection of H_0 constitutes evidence that the entropy indicator Granger-causes Brent returns, in the specific sense that it carries information about future returns that is not already embedded in the returns' own autoregressive structure.

The test statistic is a standard F -test comparing the residual sum of squares of the restricted and unrestricted models:

$$F = \frac{(\text{RSS}_r - \text{RSS}_{ur})/p}{\text{RSS}_{ur}/(T - 2p - 1)}, \quad (33)$$

where RSS_r and RSS_{ur} are the residual sums of squares of the restricted and unrestricted models respectively, p is the number of lags, and T is the effective sample size after accounting for the lag truncation. Under H_0 , this statistic follows an $F(p, T - 2p - 1)$ distribution (Granger, 1969; Hamilton, 1994).

The lag order p is selected by the Akaike Information Criterion (AIC), which is the standard choice when the primary objective is predictive accuracy rather than model parsimony (Akaike, 1974). The AIC balances goodness of fit against model complexity by penalising each additional parameter:

$$\text{AIC}(p) = \ln\left(\frac{\text{RSS}_{ur}}{T}\right) + \frac{2k}{T}, \quad (34)$$

where $k = 2p + 1$ is the total number of estimated parameters in the unrestricted model. The criterion is evaluated for every candidate lag order from $p = 1$ to a ceiling of $p_{\max} = 30$ trading days (approximately six trading weeks), and the lag order that minimises the AIC is selected for the F -test. This ceiling is deliberately conservative: it is wide enough to reveal lead-lag relationships extending beyond the short horizons typically examined in daily-frequency studies, yet tight enough to avoid the parameter proliferation that a much larger window would invite, which the AIC penalty alone might not adequately discipline in a sample of this length.

It is worth noting explicitly what Granger causality does and does not establish. The test provides evidence of incremental predictive content in a forecasting sense: lagged entropy values carry information about future returns that is not already captured by the returns' own lags. It does not establish structural or economic causation (Granger,

1969). The entropy indicator may lead returns because it captures the early stages of a distributional shift that subsequently manifests in realised price movements, or because both series respond to a common underlying driver with different speeds. The test is agnostic about the mechanism; it speaks only to the statistical predictability that an early warning system requires.

7.5 Internal Coherence of the Entropy Indicators

Before interpreting the lead-lag results, it is useful to examine whether the three entropy components, full, left-tail, and right-tail, move together or exhibit distinct dynamics. This is assessed by computing the pairwise CCF among all three indicators. If the components share a common factor, their pairwise CCFs should peak near zero lag with high correlation. If they respond to different types of shocks with different timing, the pairwise CCFs may reveal non-trivial lead-lag structure among the entropy measures themselves.

The coherence test applied here follows the internal-consistency diagnostic used in composite-indicator construction (Bujosa et al., 2013). In that framework, a well-constructed composite leading indicator should have components that are internally coherent, meaning they move together contemporaneously ($h^* \approx 0$) rather than leading or lagging one another. The expectation for the entropy indicators is similar but not identical: the full-distribution entropy is by construction a function of both tails, so it should be contemporaneously correlated with each tail component. The left-tail and right-tail entropies, however, respond to different types of shocks, downside and upside respectively, and need not be correlated with each other.

7.6 Results

The early warning battery is applied to the three rolling-window entropy indicators (full, left-tail, and right-tail KL divergence against the Normal baseline) computed at $h = 5$ over the out-of-sample period from June 2012 to April 2026. After first-differencing the entropy series, the paired sample used for the cross-correlation and Granger analyses comprises 2,167 aligned daily observations (the Brent return entering in levels).

7.6.1 Cross-Correlation Function Results

All three entropy indicators lead Brent returns: the peak cross-correlation falls at a strictly positive lag in every case and is negative throughout. The right-tail entropy peaks earliest and strongest ($h^* = +6$, $r = -0.250$), the full-distribution entropy at $h^* = +14$ ($r = -0.229$), and the left-tail entropy latest and weakest ($h^* = +16$, $r = -0.110$); all exceed the Bartlett 95 percent band of ± 0.042 in absolute value.⁴ The common negative

⁴Because each entropy indicator is itself computed from a density forecast targeted $h = 5$ days ahead, part of the measured lead is mechanical rather than discovered. The cross-correlation and Granger exercises are therefore intended to confirm that this anticipatory content is statistically detectable in realised returns, not to claim it was found by accident.

sign is economically coherent: rising distributional stress precedes weaker returns, the same channel that underlies the *Growth-at-Risk* paradigm. The association is modest in magnitude, however—the squared peak correlations imply that only about one to six percent of return variation is linearly linked to the leading signal (6.3% right tail, 5.2% full, 1.2% left).

That all three peaks occur at $h^* > 0$ is the central finding: over this sample the entropy measures behave as leading, not coincident, indicators. The CCF is only bivariate, though, and cannot rule out a common third factor driving both series; the Granger test in the next subsection controls for this in part by conditioning on the returns’ own history.

7.6.2 Granger Causality Results

The Granger causality tests reinforce the CCF findings. With the AIC search ceiling at $p_{\max} = 30$ lags, the criterion selects lag orders of 16 to 22 trading days across the three indicators.

Table 6: Granger causality test results: entropy indicators $\rightarrow$ Brent returns.

Indicator	F -statistic	p -value	Significance
Full Entropy	≈ 29.4	< 0.001	***
Left Entropy	≈ 5.2	< 0.001	***
Right Entropy	≈ 29.6	< 0.001	***

Notes: *** significant at the 1% level. Lag orders selected by AIC (search ceiling $p_{\max} = 30$) fell in the range 16–22 across the three indicators. Sample: 2,167 aligned daily observations (June 2012 – April 2026); entropy first-differenced, returns in levels. F -statistics recomputed by the author from the stored entropy series and reported to one decimal. Source: Granger (1969); lag selection via Akaike (1974).

All three entropy indicators Granger-cause Brent returns at the 1 percent level ($p < 10^{-3}$): lagged entropy carries predictive content beyond the returns’ own autoregressive structure. The F -statistics order the indicators clearly— right-tail ($F \approx 29.6$) and full-distribution ($F \approx 29.4$) entropy dominate, with the left tail a distant third ($F \approx 5.2$). This is consistent with upside stress being driven by abrupt supply-side and geopolitical shocks that the density detects early, whereas downside risk unfolds more gradually and is largely already absorbed by the returns’ own history.

These results are suggestive rather than definitive. The selected lag orders imply a moderately large model (around forty parameters), and although the sample of 2,167 observations affords ample degrees of freedom, in-sample predictability of this kind cannot be separated from possible overfitting without a formal out-of-sample forecasting exercise.

7.6.3 Internal Coherence

The pairwise CCF analysis among the three entropy indicators reveals a pattern of partial coherence, reported in Table 7.

Table 7: Internal coherence: pairwise CCF among entropy indicators.

Series X	Series Y	h^*	r at h^*	Significant
Full Entropy	Left Entropy	0	+0.748	Yes
Full Entropy	Right Entropy	0	+0.631	Yes
Left Entropy	Right Entropy	0	-0.036	No

Bartlett 95% confidence band: $\pm 1.96/\sqrt{N}$.

The left-tail and right-tail entropies are essentially uncorrelated ($r = -0.036$, $h^* = 0$, not significant), which is consistent with the interpretation that they respond to distinct types of market events. A supply-disruption shock that fattens the right tail of the return distribution does not necessarily fatten the left tail simultaneously, and a demand-contraction episode that elevates left-tail entropy need not generate upside tail activity. If this interpretation is correct, the two tail components carry largely independent information, which would explain why monitoring both separately, rather than relying on the full-distribution aggregate alone, may provide a richer early warning profile.

7.7 Synthesis and Implications

The combined evidence from the Cross-Correlation Function and the Granger causality test is consistent with the hypothesis that the entropy measures derived from the *Oil-at-Risk* framework contain early warning information about oil price movements over the sample studied. The main observations are as follows. All three entropy indicators exhibit peak cross-correlations with Brent returns at positive lags, ranging from 6 to 16 trading days, so that over this sample they behave as leading rather than contemporaneous diagnostics. The peak correlations are negative throughout, meaning that a rise in distributional stress tends to precede weaker returns, and although the associated lead times are non-trivial the implied linear explanatory power is modest, with squared peak correlations between roughly one and six percent. The Granger causality tests indicate that the predictive content of the entropy indicators is incremental: it is not replicable by the returns' own autoregressive structure alone. The right-tail entropy tends to provide the strongest and fastest signal, consistent with the abrupt, event-driven nature of supply-side shocks in crude oil markets, while the left-tail entropy provides a weaker but still statistically significant signal, consistent with the slower propagation of demand-side risks. Finally, the internal coherence analysis indicates that the left-tail and right-tail entropies respond to largely distinct phenomena and carry independent information about different types of oil market risk.

From a practical perspective, the system suggests the following operational logic over the period examined, subject to the caveat that the measured associations, while statistically detectable, are individually weak. A risk manager monitoring the *Oil-at-Risk* dashboard

observes the daily rolling entropy series decomposed by tail. A rise in right-tail entropy indicates that the conditional distribution is developing unusually heavy upside dispersion, potentially driven by geopolitical escalation or supply tightening, and this elevated stress has tended, over this sample, to precede softer realised returns within roughly one to one and a half trading weeks. A rise in left-tail entropy indicates the accumulation of downside tail risk, potentially associated with weakening demand conditions or an appreciating dollar, with a somewhat longer and weaker lead. The full-distribution entropy, which aggregates both tails, provides the broadest signal. These lead times are long enough to be of potential use in risk management, informing decisions such as adjusting hedge ratios, recalibrating margin requirements, or revising scenario assumptions for stress testing; given the modest effect sizes, however, the indicator is best read as one input among several rather than a standalone trigger. It nonetheless extends the distributional analysis of the *Oil-at-Risk* framework from a purely retrospective diagnostic toward a forward-looking surveillance tool.

8 Conclusion

This thesis built and evaluated *Oil-at-Risk*, a probabilistic tail-risk framework for daily Brent crude returns that adapts the *Growth-at-Risk* paradigm of Adrian et al. (2019) to the energy domain: quantile regressions on geopolitical and macroeconomic conditioning variables, estimated by direct multi-step forecasting; a continuous Johnson S_U conditional density fitted by Minimum Distance Estimation at each origin; and Value-at-Risk, Expected Shortfall, Kullback–Leibler divergence, and skewness read from that density. Across the out-of-sample period the fitted densities were statistically adequate (the PIT did not reject uniformity), all three specifications passed the Kupiec coverage test at roughly 97–98 percent, and none dominated on predictive score, so the parsimonious standard model was retained with the CAViaR variants kept as shorter-horizon complements.

The central conclusion is that these distributional outputs can be assembled into a genuine early warning system. The entropy indicator, the tail-decomposed Kullback–Leibler divergence of the conditional density, led Brent returns by 6 to 16 trading days over the sample and Granger-caused them at the 1 percent level, so the framework does not merely describe risk after the fact but anticipates it. The signal responds to both sources of tail activity that matter in this market: the geopolitical shocks that enter through the conditioning variables, and the broader movements in market volatility, including speculative positioning and directional-dispersion dynamics, that reshape the conditional density whatever their origin. That this anticipatory content survives at *daily* frequency, and in an asset as volatile as crude oil, is the result that matters: it is precisely in a high-frequency, high-volatility setting that the backward-looking, Gaussian risk measures of standard practice are least reliable and an early, tail-specific warning is most valuable.

The lead times observed here are long enough to be operationally meaningful, giving a risk manager a window to adjust hedges and margins before stress is realised rather than after.

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A Optimisation Method and Hypotheses

In standard OLS, because the squared loss function creates a smooth parabola, taking the derivative and setting it to zero yields a closed-form algebraic solution: $\hat{\beta} = (X^\top X)^{-1} X^\top Y$. However, the absolute value in the check function creates a sharp “V” shape at zero. This non-differentiable corner means that standard first-order conditions cannot be applied. Instead, the optimisation is transformed into a Linear Programming (LP) problem.

To eliminate the absolute piecewise nature of the function, every residual is split into two strictly non-negative directional buckets: u_i^+ (the under-prediction bucket, measuring the magnitude by which the actual value exceeded the prediction) and u_i^- (the over-prediction bucket, measuring the magnitude by which the prediction exceeded the actual value). Every residual can now be expressed as a linear combination $u_i = u_i^+ - u_i^-$. Substituting this into the regression equation ($y_i - x_i^\top \beta = u_i^+ - u_i^-$) and rearranging defines the structural constraints for the optimisation algorithm:

$$x_i^\top \beta + u_i^+ - u_i^- = y_i. \quad (35)$$

Applying this transformation removes the absolute value operator from the objective function. The minimisation problem is expressed in a standard primal Linear Programming format:

$$\min_{\beta, u^+, u^-} \sum_{i=1}^n [\tau u_i^+ + (1 - \tau) u_i^-] \quad \text{subject to} \quad x_i^\top \beta + u_i^+ - u_i^- = y_i, \quad u_i^+, u_i^- \geq 0. \quad (36)$$

By formulating the problem this way, statistical software can solve the regression using efficient LP algorithms, primarily the Barrodale–Roberts modified simplex algorithm (Barrodale and Roberts, 1973) for smaller datasets, which searches for the minimum along the geometric edges of the data intersections, or Frisch–Newton interior-point methods (Portnoy and Koenker, 1997) for large datasets, which cut through the feasible space using a logarithmic barrier to ensure positivity.

For $\hat{\beta}_\tau$ to be a consistent estimator, the model relies on several core hypotheses. Linearity of conditional quantiles requires that the conditional quantile has a linear relationship with the regressors, such that $Q_\tau(Y | X) = X^\top \beta(\tau)$. Strict exogeneity requires that the conditional quantile of the error term, given X , is zero: $Q_\tau(u | X) = 0$. No perfect multicollinearity requires that the matrix $\mathbb{E}[XX^\top]$ is finite, positive-definite, and invertible. Positive density requires that the conditional density function of the error term around zero is strictly positive and finite ($0 < f_{u|X}(0 | X) < \infty$). Because precision in quantile regression depends on the density of observations near the estimated quantile, this condition ensures the estimator is well-identified.

B Quantile Regression Diagnostics: Full Technical Development

This appendix presents the methodological foundation of each diagnostic test applied to the specification. For each test, the statistical framework, null hypothesis, test statistic, and asymptotic distribution are stated. Throughout, $\rho_\tau(u) = u[\tau - \mathbf{1}(u < 0)]$ denotes the check (pinball) function introduced in Section 5.1, and k denotes the number of slope parameters in the specification.

B.1 Goodness-of-Fit: Koenker–Machado $R^1(\tau)$

Unlike Ordinary Least Squares, quantile regression does not minimise a quadratic objective and therefore possesses no natural coefficient of determination. The local goodness-of-fit measure introduced by Koenker and Machado (1999) resolves this by defining a pseudo- R^2 as the proportional reduction in the asymmetrically weighted sum of absolute errors relative to an intercept-only null model:

$$R^1(\tau) = 1 - \frac{\sum_{t=1}^T \rho_\tau(y_t - x'_t \hat{\beta}(\tau))}{\sum_{t=1}^T \rho_\tau(y_t - \hat{\xi}_\tau)}, \quad (37)$$

where the numerator is the optimised value of the check-loss objective for the full specification and the denominator is the same objective evaluated at the intercept-only model, whose solution $\hat{\xi}_\tau$ is the unconditional sample τ -quantile. Because both sums are strictly non-negative and the full model can only reduce the objective relative to the restricted one, the metric is bounded in $[0, 1]$ and is interpreted locally at each τ : a value of $R^1(\tau) = 0.10$ at $\tau = 0.05$ indicates that the regressors reduce the weighted objective by ten percent relative to the unconditional quantile in that tail. Koenker and Machado (1999) establish that the statistic is consistent and that, under standard regularity conditions,

$$T \cdot V(\tau) \cdot R^1(\tau) \xrightarrow{d} \chi_k^2, \quad (38)$$

where k is the number of slope parameters and $V(\tau)$ is a nuisance scale that depends on the sparsity of the error distribution at τ , that is, on the reciprocal of the conditional error density evaluated at the τ -quantile. The dependence on sparsity is the same feature that governs the asymptotic variance of the quantile-regression coefficients themselves: precision deteriorates wherever the data are thin around the fitted quantile.

B.2 Wald Test for Equality of Coefficients across Quantiles

A foundational identifying assumption of the *Oil-at-Risk* framework is that the effect of the conditioning variables on the return distribution is heterogeneous across quantiles. If the slope coefficients were identical for every τ , the distribution would shift in parallel and quantile regression would collapse to a reparameterisation of OLS. The Wald test formalises this by testing coefficient constancy across the m estimated quantiles jointly (Koenker and Bassett, 1978). Let

$$\hat{\zeta} = \left(\hat{\beta}(\tau_1)', \hat{\beta}(\tau_2)', \dots, \hat{\beta}(\tau_m)' \right)' \quad (39)$$

be the $mp \times 1$ stacked parameter vector across all m quantiles, where p is the dimension of β at each quantile. A full-rank $q \times mp$ restriction matrix R is constructed to extract the relevant slope coefficients and impose equality across quantile pairs, so that the null reads $H_0: R\zeta = r$, with r the $q \times 1$ vector of hypothesised values (typically zero for difference restrictions). The Wald statistic is

$$T_n = n(R\hat{\zeta} - r)^\top [R\hat{V}_n R^\top]^{-1} (R\hat{\zeta} - r) \xrightarrow{d} \chi_q^2, \quad (40)$$

where $\hat{V}_n$ is the joint asymptotic covariance matrix of $\hat{\zeta}$. Because the parameters at different quantiles are estimated from the same time series, their sampling errors are correlated across quantiles, so the off-diagonal blocks of $\hat{V}_n$ are non-zero and cannot be ignored. Analytical sparsity estimators are moreover unreliable in the presence of temporal dependence. Accordingly, $\hat{V}_n$ is constructed by a moving block bootstrap: contiguous blocks of observations of length l , chosen to span the autocorrelation horizon of the data, are resampled, and the full quantile system is re-estimated on each replication, so that the resampled blocks preserve the serial dependence that an i.i.d. bootstrap would destroy.

B.3 Quantile Crossing Inspection

A necessary condition for any valid conditional distribution function is that the conditional quantile function be monotonically non-decreasing in τ . Formally, for any two levels $\tau_1 < \tau_2$ and any observation t ,

$$\hat{q}_t(\tau_1) \leq \hat{q}_t(\tau_2) \quad \forall t \in \{1, \dots, T\}. \quad (41)$$

Violations of this condition, known as quantile crossings, arise when the estimated linear quantile planes intersect within the covariate support, typically as a consequence of over-parameterisation, collinear regressors, or an insufficient number of observations relative to the complexity of the model. Crossings are detected by computing, for each observation t , the ordered sequence of fitted values across all targeted quantiles and checking for any inversion of that ordering. The proportion of crossing observations and the magnitude of

the maximal inversion are reported as informal diagnostics; a specification with systematic crossings in the interior of the covariate support is treated as misspecified.

B.4 Stationarity: Quantile Autoregressive Unit Root Test

Standard unit-root tests evaluate the null of a unit root against the alternative of global mean reversion, implicitly assuming a single autoregressive parameter across the whole distribution. Koenker and Xiao (2004) relax this by permitting quantile-specific autoregressive dynamics, so that a series may be stationary at some quantiles while behaving near a unit root at others. The test is implemented through the auxiliary error-correction regression

$$\Delta y_t = \mu(\tau) + \rho(\tau) y_{t-1} + \sum_{j=1}^p \gamma_j(\tau) \Delta y_{t-j} + \epsilon_t(\tau), \quad (42)$$

where the lag order p is selected by an information criterion applied to the augmented Dickey–Fuller pre-regression. The null $H_0: \rho(\tau) = 0$ posits a quantile-specific unit root and is tested against the one-sided alternative $H_1: \rho(\tau) < 0$. The statistic $t_{\text{QAR}}(\tau) = \hat{\rho}(\tau)/\widehat{\text{se}}(\hat{\rho}(\tau))$ is obtained from the quantile objective and does not follow a standard normal distribution under the null. Koenker and Xiao (2004) show that its limiting distribution is

$$t_{\text{QAR}}(\tau) \xrightarrow{d} \frac{f(F^{-1}(\tau))}{\sqrt{\tau(1-\tau)}} \cdot \frac{\int_0^1 W(s) dW(s)}{\left(\int_0^1 W(s)^2 ds\right)^{1/2}}, \quad (43)$$

where $W(s)$ is a standard Brownian motion on $[0, 1]$, f is the marginal density of the innovations, and $F^{-1}(\tau)$ is the corresponding population quantile. The right-hand side is the standard Dickey–Fuller functional scaled by the sparsity factor $f(F^{-1}(\tau))/\sqrt{\tau(1-\tau)}$, which varies with τ and renders the ordinary Dickey–Fuller critical values inapplicable. Quantile-specific critical values are therefore obtained by simulation following Koenker and Xiao (2004).

B.5 Dynamic Quantile Test

The Dynamic Quantile test of Engle and Manganelli (2004) evaluates whether the sequence of quantile violations behaves as a martingale difference sequence, i.e. whether breaches are unpredictable from past information. The centred hit sequence is

$$\text{Hit}_t(\tau) = \mathbf{1}(y_t < \hat{q}_t(\tau)) - \tau, \quad (44)$$

which has mean zero by construction under correct specification. The null hypothesis is

$$H_0: \mathbb{E}[\text{Hit}_t(\tau) \mid \Omega_{t-1}] = 0, \quad (45)$$

so that the violation process is orthogonal to all past information Ω_{t-1} . To test this, the hits are projected by OLS onto an instrument matrix Z_t comprising a constant, L lagged hits $\text{Hit}_{t-1}(\tau), \dots, \text{Hit}_{t-L}(\tau)$, and the lagged fitted quantile $\hat{q}_{t-1}(\tau)$:

$$\text{Hit}_t(\tau) = z_t' \delta + \eta_t. \quad (46)$$

Engle and Manganelli (2004) show that under the null the Wald form of the statistic,

$$DQ(\tau) = \frac{\text{Hit}' Z (Z' Z)^{-1} Z' \text{Hit}}{\tau(1 - \tau)} \xrightarrow{d} \chi_q^2, \quad (47)$$

is asymptotically chi-squared with q degrees of freedom equal to the column dimension of Z . The asymptotically equivalent $N \cdot R^2$ form of the Lagrange-multiplier statistic converges to the same distribution and is reported for comparability. The test is applied separately at each targeted quantile; a joint test across quantiles is constructed by stacking the instrument matrices and adjusting the degrees of freedom accordingly. As explained in Section 5.5.3, the CAViaR specifications are designed to pass this test by construction, since their regressors are built precisely from the lagged breach information that appears in Z_t .

C Minimum Distance Estimator

Two algorithmic choices materially affect the reliability of the solution. First, the initial guess Θ_0 strongly influences the speed and likelihood of convergence. The initialisation sets ξ at the predicted median $\hat{q}_{t+h}(0.50)$, λ at the interquartile range $\hat{q}_{t+h}(0.75) - \hat{q}_{t+h}(0.25)$, and γ and δ at zero and one, respectively. This encodes a symmetric, moderately heavy-tailed prior centred on the regression's median forecast, a sensible starting point that the optimiser then refines as it absorbs the asymmetry embedded in the outer quantiles. Second, convergence tolerances must be tight enough to produce an accurate fit but loose enough to avoid excessive iterations on nearly flat regions of the objective (sum-of-squares) surface; a function tolerance of 10^{-10} has proven adequate across the full out-of-sample experiment.⁵

The procedure is repeated independently for every forecast origin in the rolling window, yielding a time series of fitted parameter vectors $\{\hat{\Theta}_t\}_{t \in W}$ and, through them, a time series of continuous conditional densities. Each density is anchored to the information

⁵The objective in (17) weights the $K = 7$ quantile residuals equally in an unweighted L_2 sum. Equal weighting is deliberate: it treats the fit to each targeted quantile as equally informative about the shape of the density and avoids importing an ad hoc weighting scheme that would itself require justification. Because the problem matches $K = 7$ target quantiles to a four-parameter family, it is over-identified with three residual degrees of freedom, so the fit is a genuine estimation rather than exact interpolation; the fitted density is not forced through every point and the residual pattern is itself a diagnostic of misspecification. A density- or spacing-weighted variant is a natural robustness check but is not pursued here.

set available at its forecast origin: it inherits the conditioning embedded in the quantile forecasts and imposes no cross-sectional pooling across dates. The result is a sequence of distributions that evolves with the geopolitical and macroeconomic environment, fattening its left tail when the Geopolitical Risk Index spikes and narrowing it when conditions normalise, which is the object the *Oil-at-Risk* framework requires. An in-sample example of a fitted conditional density is displayed as a three-dimensional surface in Figure 3.

D The Johnson S_U Distribution

The parametric family used to interpolate the quantile skeleton in Section 5 must satisfy two requirements. First, it must be flexible enough to accommodate the stylised facts of crude oil returns: excess kurtosis, asymmetric tails, and time-varying skewness whose direction shifts as geopolitical conditions evolve. When a supply disruption threatens, the right tail of the return distribution fattens as the market prices in the possibility of a sharp price spike; when a demand collapse materialises, the left tail fattens instead. A distribution that imposes symmetry, or that controls tail weight through a single parameter, cannot track these asymmetric rotations. Second, its cumulative distribution function must admit a closed-form inverse, because the estimation procedure evaluates the theoretical quantile function many thousands of times.

The Johnson S_U distribution satisfies both. A random variable Y follows a Johnson S_U if

$$Y = \xi + \lambda \sinh\left(\frac{Z - \gamma}{\delta}\right), \quad Z \sim \mathcal{N}(0, 1), \quad (48)$$

where ξ is the location parameter, $\lambda > 0$ is the scale, γ governs skewness, and $\delta > 0$ governs tail weight. The quantile function follows by replacing Z with $\Phi^{-1}(p)$:

$$Q(p) = \xi + \lambda \sinh\left(\frac{\Phi^{-1}(p) - \gamma}{\delta}\right). \quad (49)$$

This closed-form quantile function is the decisive computational advantage: it evaluates rapidly, involves no numerical root-finding, and makes the Minimum Distance Estimation step as fast as a simple least-squares problem.

The economic appeal of the S_U family becomes clear when one considers the alternative. The Generalised Error Distribution, widely used in GARCH modelling (Nelson, 1991), controls tail weight through a single shape parameter but imposes symmetry: it cannot distinguish between a left-skewed distribution during a demand collapse and a right-skewed one during a supply panic. The Johnson S_U provides independent control over skewness and kurtosis through the two separate shape parameters γ and δ . In the limit $\gamma \rightarrow 0$, $\delta \rightarrow \infty$, the S_U nests the Gaussian as a special case, so that tranquil periods are accommodated without forcing heavy tails where none exist. The original *Growth-at-Risk*

framework of Adrian et al. (2019) employs the Azzalini skewed- t distribution for this step; the S_U is a more finance-oriented alternative that has been used to model asset returns in portfolio management and Value-at-Risk contexts, with comparable flexibility and a simpler quantile-function inversion.

D.1 Kullback–Leibler Divergence: Definition and Decomposition

Kullback–Leibler divergence (Kullback and Leibler, 1951) provides a global measure of distributional distance. Given a conditional forecast density $p(x)$ and a reference density $q(x)$,

$$D_{\text{KL}}(p \parallel q) = \int_{-\infty}^{\infty} p(x) \ln \left(\frac{p(x)}{q(x)} \right) dx. \quad (50)$$

The divergence is always non-negative and equals zero if and only if the two densities are identical. Its asymmetry is economically meaningful: $D_{\text{KL}}(p \parallel q)$ heavily penalises the reference density q when it assigns near-zero probability to events that the conditional density p considers plausible—in risk-management terms, the dangerous scenario in which the reference model says a large loss is essentially impossible while the conditional forecast says it has non-negligible probability.

Through the lens of information theory (Shannon, 1948), the divergence is the expected excess surprise when q is used to describe a world actually governed by p :

$$D_{\text{KL}}(p \parallel q) = \mathbb{E}_p[-\ln q(x)] - \mathbb{E}_p[-\ln p(x)]. \quad (51)$$

The first term is the cross-entropy of q under p ; the second is the entropy of p itself. The divergence measures the extra information, in nats, needed to describe the true state of the world when the wrong model is assumed.

Following Adrian et al. (2019), the divergence is decomposed into left-tail, centre, and right-tail components by restricting the integration domain. With $\tau = 0.95$ as the default tail mark (isolating the extreme 5 percent of each tail, rather than the median split used in the original *Growth-at-Risk* framework, which is better suited to quarterly GDP growth than to daily commodity returns),

$$D_{\text{KL}}^{\text{Left}}(p \parallel q) = \int_{-\infty}^{q_{1-\tau}} p(x) \ln \left(\frac{p(x)}{q(x)} \right) dx, \quad (52)$$

$$D_{\text{KL}}^{\text{Right}}(p \parallel q) = \int_{q_\tau}^{\infty} p(x) \ln \left(\frac{p(x)}{q(x)} \right) dx. \quad (53)$$

Two reference baselines are employed. The first is a Normal density calibrated to the historical mean and standard deviation, whose divergence measures the departure from Gaussianity and functions as an early warning: a rapid increase signals that the market’s return distribution is moving away from the well-behaved state in which traditional

risk metrics are reliable.⁶ The second is the unconditional Johnson S_U density fitted by Maximum Likelihood to the historical returns, which is the baseline used in the original Vulnerable Growth framework and is shown in Figure 11. This divergence is more demanding, because the unconditional S_U already accommodates skewness and heavy tails; a high divergence against it signals a genuine shift in the conditional distribution, not merely a departure from Gaussianity.

E Out-of-Sample Forecast Evaluation Tests

This appendix documents every test used to evaluate the direct-forecasting models. Two conventions hold throughout. First, a hit (or breach) is defined under a unified directional rule: a hit occurs whenever the realised value falls strictly below the forecast, $y_{t+h} < \hat{y}_{t+h|t}(\tau)$, so that for any quantile the model should be breached exactly $\tau \times 100\%$ of the time. Second, all coverage tests are run out-of-sample on a fixed 1,000-day rolling window, re-estimated at every forecast origin.

E.1 The Rolling-Window Experiment

Every statistic is computed on genuinely out-of-sample forecasts generated by a fixed-width rolling window. For each forecast origin t in the test span the model is trained on the 1,000 observations in the half-open interval $[t - 1000, t)$, a single h -step-ahead prediction of Y_{t+h} is produced, and the window rolls forward one day. Because the realised target Y_{t+h} never enters the training window, there is no load-bearing look-ahead. The reported figures at $h = 5$, $\tau = 0.95$ are averages over hundreds of origins: 864 out-of-sample forecasts for the benchmark (2 January 2023 to 23 April 2026) and 517 for the CAViaR variants (2 January 2023 to 20 April 2026), the difference reflecting the extra rows consumed by constructing the breach regressors.

E.2 Pinball Loss and Scoring

In conditional-mean forecasting the mean-squared error is the natural default, but symmetry is precisely the wrong property for a quantile, where over- and under-prediction must be penalised by different amounts. The asymmetric pinball (check) loss supplies that asymmetry:

$$L_\tau(e) = \begin{cases} \tau e & \text{if } e \geq 0, \\ (\tau - 1) e & \text{if } e < 0, \end{cases} \quad e = y_{t+h} - \hat{y}(\tau), \quad (54)$$

⁶The Kullback–Leibler divergence of a heavy-tailed p against a Gaussian q is dominated by the region where $q \rightarrow 0$, so this measure behaves primarily as a detector of tail-fatness (excess kurtosis) rather than of location or mild skewness; this is a feature for early warning, since tail-fattening is precisely the object of interest, but it should be read as a kurtosis-sensitive signal rather than a generic “distance”. The tail integrals are evaluated on a finite grid, and the truncation range is set wide enough that the reported divergences are numerically stable across the sample.

which is always non-negative and can be written compactly as $L_\tau(e) = e[\tau - \mathbf{1}(e < 0)]$. A scoring rule is *strictly proper* if its expectation is minimised if and only if the issued forecast equals the true value of the targeted functional. The pinball loss is the unique (up to affine transformation) strictly proper scoring rule for a quantile (Gneiting and Raftery, 2007), which is the formal counterpart of the statement that the quantile is an *elicitable* functional. The practical consequence is strong: if model A attains a lower out-of-sample pinball loss than model B at quantile τ , then A is a better estimator of that τ -quantile, and the comparison requires no distributional assumption. This is what licenses ranking competing specifications directly by their realised loss.

E.3 Kupiec Proportion-of-Failures Test

Before considering the timing of breaches, the simplest question is whether the τ -quantile forecast is breached with the correct *frequency*. The Kupiec proportion-of-failures test (Kupiec, 1995) checks whether the empirical hit rate matches the nominal level; it is unconditional because it judges frequency alone and is silent about clustering. Let T be the number of out-of-sample forecasts and x the number of hits, with empirical rate $\hat{\pi} = x/T$. Under the null the breach indicator is Bernoulli with the nominal probability, so

$$H_0: \mathbb{E}[I_{t+h}] = \tau \quad \text{vs.} \quad H_1: \mathbb{E}[I_{t+h}] \neq \tau, \quad (55)$$

and the likelihood-ratio statistic contrasts the restricted likelihood (probability fixed at the nominal level) with the unrestricted one (probability free at $\hat{\pi}$):

$$LR_{UC} = -2 \ln \left[\frac{(1-\tau)^{T-x} \tau^x}{(1-\hat{\pi})^{T-x} \hat{\pi}^x} \right] \xrightarrow{d} \chi_1^2. \quad (56)$$

By Wilks' theorem the statistic is asymptotically chi-squared with degrees of freedom equal to the difference in the number of free parameters between the two likelihoods, here one (the free rate) minus zero (the fixed target), hence a single degree of freedom. Because the hit count is binomial, the central limit theorem also supplies a normal-approximation band around the nominal rate, $\tau \pm z\sqrt{\tau(1-\tau)/T}$, which is the envelope drawn in the coverage-convergence plots. As discussed in the main text, the overlapping structure of the h -step direct forecasts inflates the variance of the breach count but not its mean, so this unconditional-frequency assessment remains valid whereas a naive independence-based conditional-coverage test would be size-distorted; the latter is therefore not reported.

E.4 Diebold–Mariano Test of Equal Predictive Accuracy

The coverage test certifies each model in isolation; the Diebold–Mariano test (Diebold and Mariano, 1995) compares two models head-to-head. Let $L_{1,t}$ and $L_{2,t}$ be the element-wise tick losses of two competing forecasts and $d_t = L_{1,t} - L_{2,t}$ their differential. The null of

equal predictive accuracy is $H_0: \mathbb{E}[d_t] = 0$. The statistic standardises the mean differential by a heteroskedasticity- and autocorrelation-consistent (HAC) long-run variance:

$$DM = \frac{\bar{d}}{\sqrt{\widehat{LRV}/P}}, \quad \widehat{LRV} = \gamma_0 + 2 \sum_{k=1}^{h-1} \gamma_k \xrightarrow{d} t_{P-1}, \quad (57)$$

where γ_k is the sample autocovariance of d_t at lag k and P is the number of paired forecast origins. Two implementation choices follow the theory. The HAC bandwidth is set to $h - 1$, because the errors of an h -step direct forecast follow an $\text{MA}(h - 1)$ process and therefore have exactly $h - 1$ non-zero autocovariances; truncating the kernel there is efficient and consistent. The reference distribution is taken as Student- t with $P - 1$ degrees of freedom rather than the standard normal, a small-sample correction. A negative and significant $\bar{d}$ indicates that the first model carries the lower average loss; a value insignificantly different from zero indicates that the two forecasts are statistically indistinguishable in accuracy, in which case parsimony is the natural tie-breaker.

F Graphical Annex

The figures referenced throughout the text are collected here. All figures are the author’s own elaboration unless otherwise stated in the caption.

Figure 1: Evolution of the European Brent crude oil spot price over the sample period. Source: U.S. Energy Information Administration (via FRED).

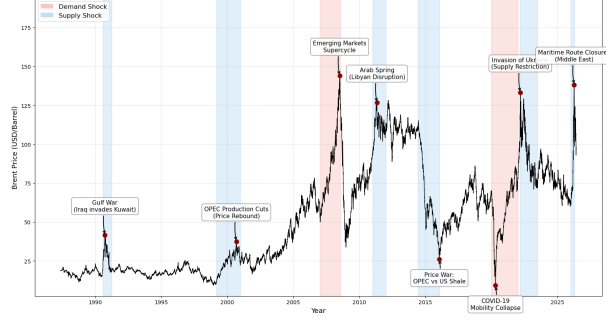
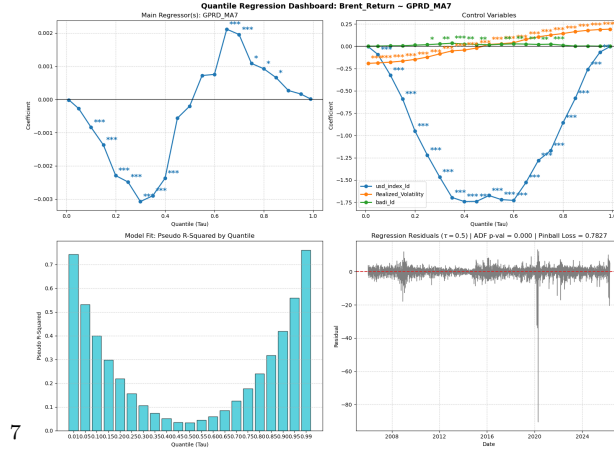


Figure 2: In-sample quantile regression coefficient paths across τ for specification (5), with the OLS benchmark for comparison. Source: author’s estimation.



⁷This in-sample specification enters the realised-volatility regressor at lag $t - 1$ rather than contemporaneously, for the reason discussed in Section 5.2: because RV_t shares its date with the target, using the one-period lag both removes the look-ahead (simultaneity) problem and prevents a function of the dependent variable from entering the right-hand side and distorting the quantile-regression coefficients.

Figure 3: Three-dimensional representation of the in-sample fitted Johnson S_U conditional density across forecast origins. Source: author's estimation.

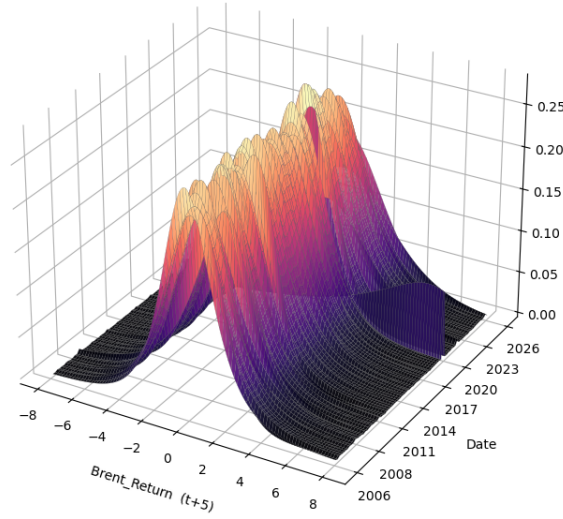


Figure 4: Realised Brent returns against the forecasted conditional quantile envelope, out-of-sample, $h = 5$. Source: author's estimation.

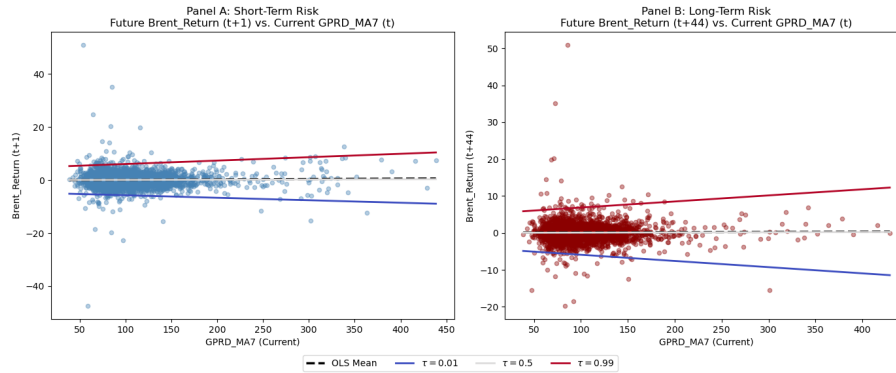


Figure 5: Average rolling out-of-sample pinball loss as a function of the forecast horizon h , for the Standard (QReg), CAViaR-indicator, and CAViaR-severity specifications at $\tau \in \{0.05, 0.5, 0.95\}$. A loss profile that rises with the horizon signals overfitting; a flat profile signals captured structural signal. Source: author's estimation.

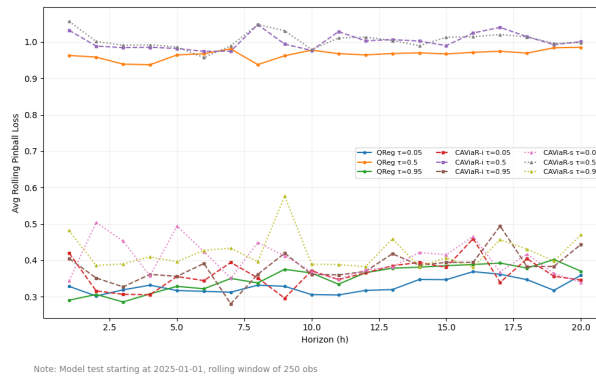


Figure 6: Standard model: in-sample quantile bounds, breaches, and fallout (magnitude of exceedance) at $h = 5$, including the behaviour of the 1%–99% tunnel during the 2020 demand collapse. Source: author's estimation.

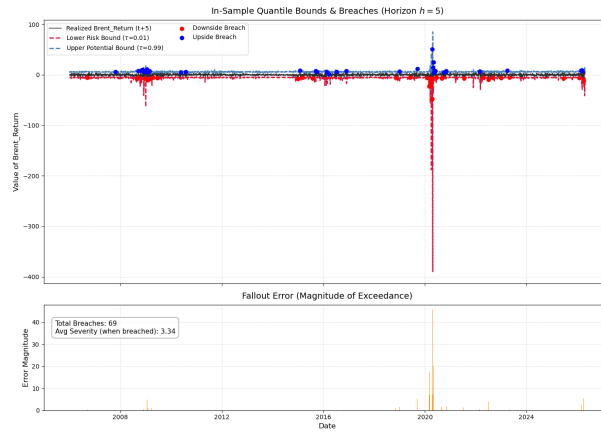


Figure 7: CAViaR-indicator model: in-sample quantile bounds, breaches, and fallout at $h = 5$. Source: author's estimation.

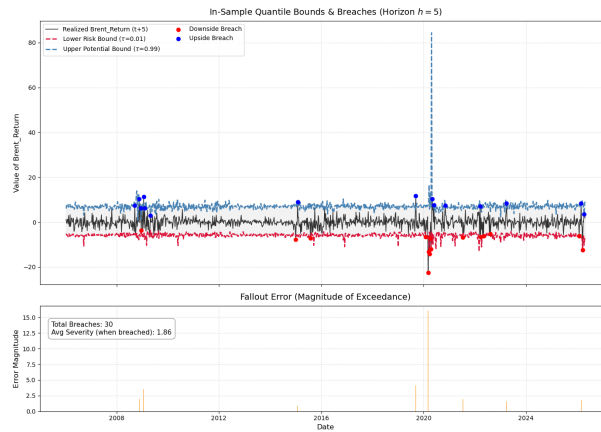


Figure 8: CAViaR-severity model: in-sample quantile bounds, breaches, and fallout at $h = 5$. Source: author's estimation.

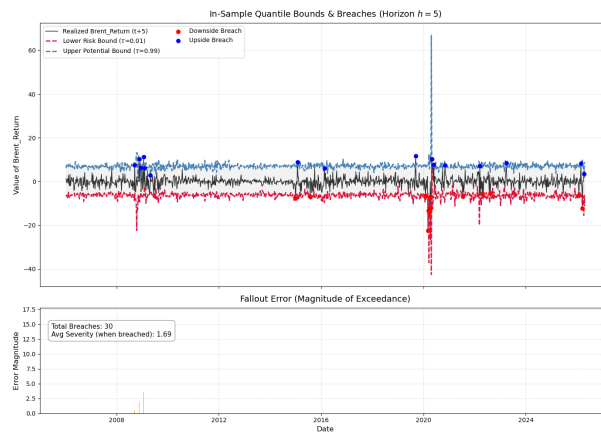


Figure 9: Unconditional (Kupiec) coverage-convergence plot at $\tau = 0.95$, $h = 5$, for the standard model: the cumulative empirical breach rate converges to the nominal target within the 95% confidence band. Source: author's estimation.

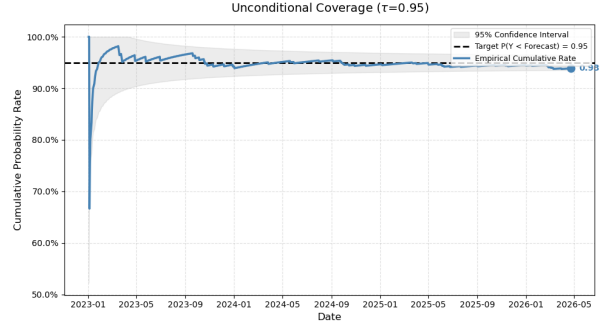


Figure 10: In-sample forecasted values during a stressed geopolitical episode (Iran-related escalation). Source: author's estimation.

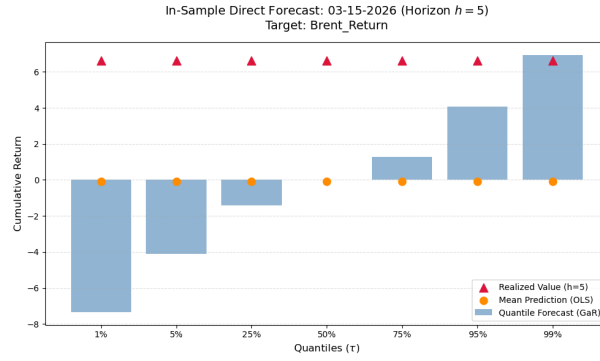


Figure 11: Unconditional Johnson S_U density fitted by Maximum Likelihood to historical Brent returns, used as a KL baseline. Source: author's estimation.

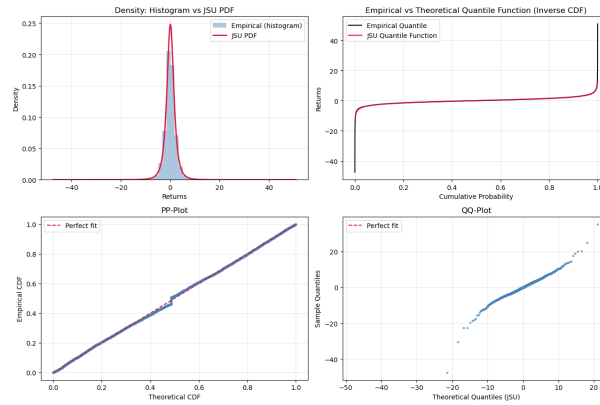


Figure 12: Probability Integral Transform histogram for the Johnson S_U conditional densities, $h = 5$, with the Kolmogorov–Smirnov statistic and p -value. Source: author’s estimation.

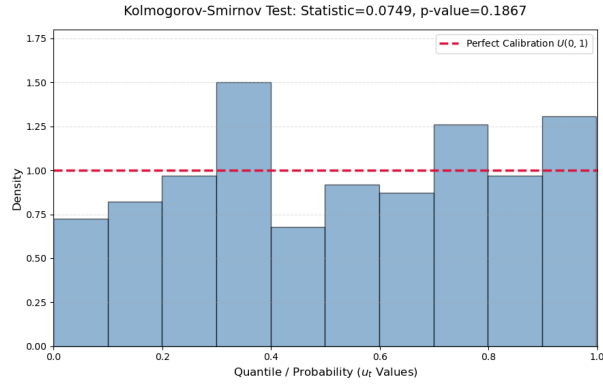


Figure 13: Probability Integral Transform diagonal plot (empirical vs. theoretical CDF). Source: author’s estimation.

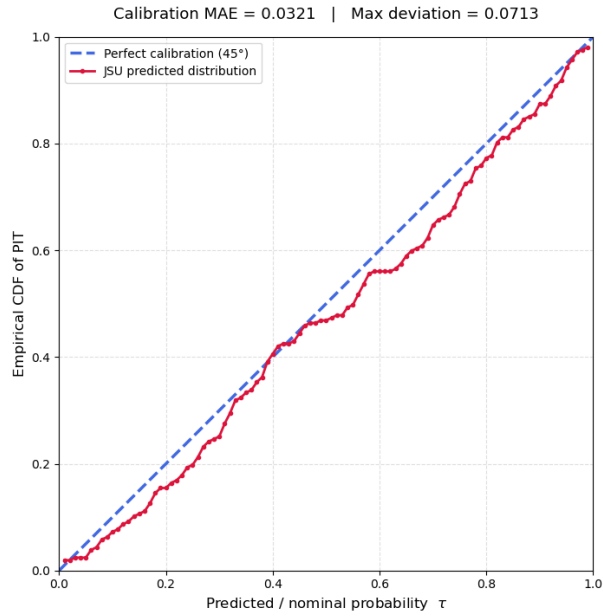


Figure 15: Kullback–Leibler entropy of the conditional density against the unconditional Johnson S_U baseline, expanding window. Source: author’s estimation.

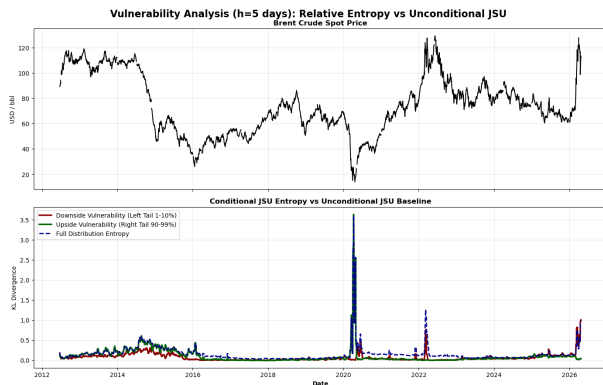


Figure 14: Kullback–Leibler entropy of the conditional density against the Normal baseline, expanding window. Source: author’s estimation.

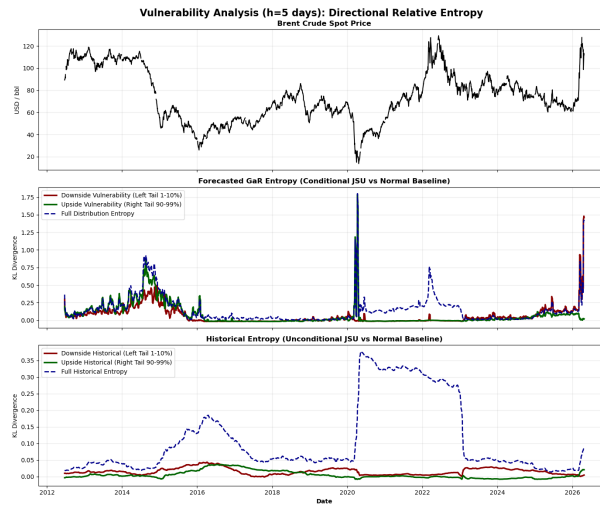


Figure 16: Conditional (*Oil-at-Risk/Growth-at-Risk*) and unconditional (historical-simulation) Value-at-Risk over the out-of-sample period, $h = 5$, 97.5% confidence. The conditional measure requires no time-scaling; the unconditional benchmark uses a four-year window with $\sqrt{5}$ scaling. Source: author’s estimation.

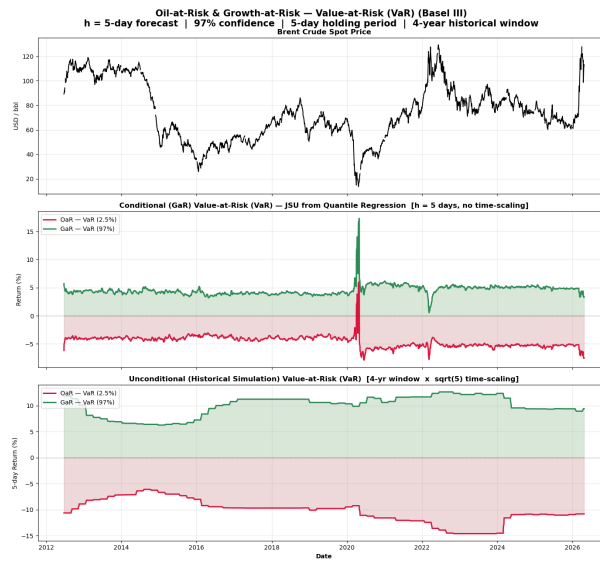


Figure 17: Conditional (*Oil-at-Risk*/*Growth-at-Risk*) and unconditional (historical-simulation) Conditional Value-at-Risk (Expected Shortfall) over the out-of-sample period, $h = 5$, 97.5% confidence. Source: author's estimation.

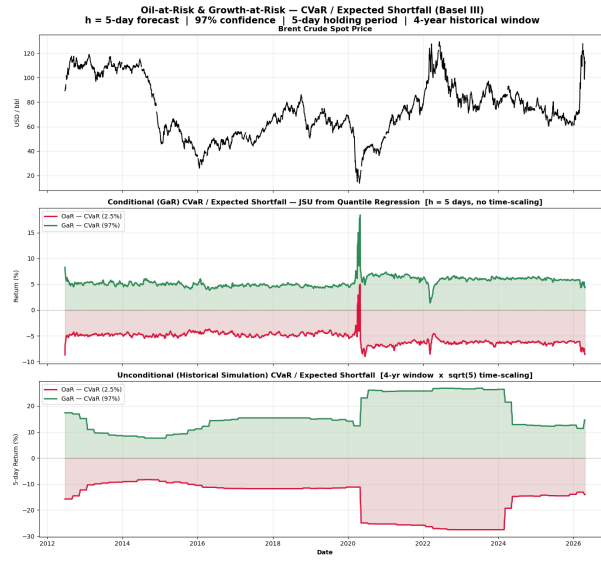


Figure 18: Cross-Correlation Function between the entropy indicators and Brent returns, with Bartlett 95% confidence bands. Source: author's estimation.

